CHAPTER 1
THE REAL NUMBER SYSTEM

1.0 Introduction

The set of **natural numbers** (or the counting numbers) is the set

$N = \{1, 2, 3, \ldots\}$.

This set is **closed** under + and $\times$, but not closed under $-$ and $\div$.

The inclusion of zero (0) gives us the set

$W = \{0, 1, 2, 3, \ldots\}$,commonly called the set of **whole numbers**. The set N may be extended to include all the negative whole numbers to give the set

$Z = \{\ldots -3, -2, -1, 0, 1, 2, 3, \ldots\}$, which is called the set of **integers.** This set Z is **closed** under +, $\times$ and $-$, but not closed under $\div$. A further extension of this set to include all fractions, both proper and improper, gives the set

$Q = \{ \frac{a}{b} : a, b \in Z, b \neq 0 \}$.

This is called the set of **Rational Numbers**. Q is closed under all the arithmetic operations +, $\times$, $-$, $\div$. In the case of closure under $\div$, division by 0 must be avoided, as the result is either indeterminate (%) or infinity (∞).

Note that

(i) $0 \in Q$, since $\frac{o}{b}$, $b \neq o$ is 0.

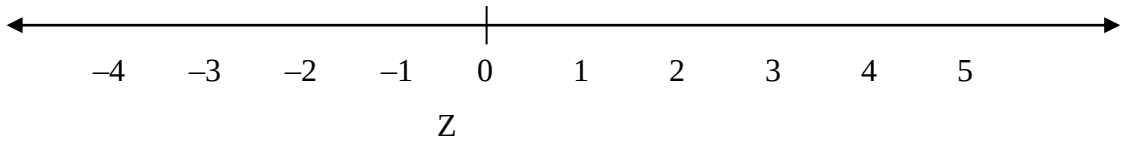
(ii) $Z \subset Q$, since $\frac{a}{1} = a$, for all $a \in Z$.

(iii) It is clear that

$N \subset Z \subset Q$.

Graphical Representation

The above number systems may be represented graphically as follows:



We may identify the subsets of Z as follows:

$Z^- = \{\ldots -3, -2, -1\}$, called the **negative integers**; and

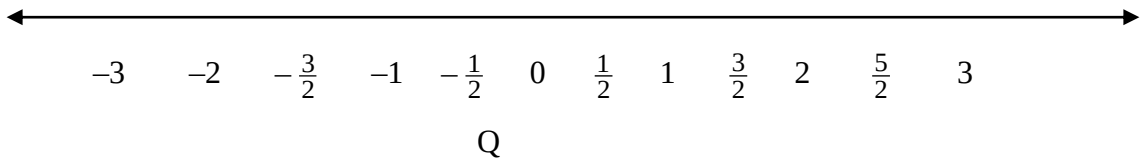
$Z^+ = \{1, 2, 3, \ldots\}$, called the positive integers. The set

$\{0\} \cup Z^+ = \{0, 1, 2, 3, \ldots\}$ is called the set of non-negative integers.

Note that

$Z = Z^- \cup \{0\} \cup Z^+$.

The rational numbers may be represented by the straight line



Every fraction whether proper or improper may be represented by a point on this line.

The Set of Irrational Numbers

There are numbers that do not belong to Q, that is, they cannot be expressed in the form $\frac{a}{b}$, where a, $b \in Z$. There are points on the number line which correspond to these numbers. The set of such numbers is called the set of Irrational Numbers. Examples of Irrational Numbers are:

$\sqrt{2}, \sqrt{3}, \pi$, e, etc.

Example

Prove that $\sqrt{2}$ is an irrational number.

Let us assume that $\sqrt{2}$ is rational. Then

$\sqrt{2} = \frac{a}{b}$, a, $b \in Z$, $b \neq o$, and such that $\frac{a}{b}$ is in its least form.

Then

$$2 = \frac{a^2}{b^2}$$

$\Rightarrow \quad a^2 = 2b^2$

$\Rightarrow$ a^2 is an even number

$\therefore$ a is even

Let $a = 2c$, $c \in Z$.

Then $(2c)^2 = 2b^2$

i.e. $4c^2 = 2b^2$

$\Rightarrow \quad b^2 = 2c^2$

$\Rightarrow$ b^2 is even $\Rightarrow$ b is even.

Since a is even and b b is even, it implies that $\frac{a}{b}$ is not in its least form. This contradiction is because of the false premise that $\sqrt{2}$ is rational. Therefore $\sqrt{2}$ is irrational.

Exercise

Prove that $\sqrt{3}$ is irrational.

Note

(i) Every rational number may be expressed either as a terminating decimal or as a recurring decimal.

E.g. $\frac{1}{2} = 0.5$,

$\frac{2}{5} = 0.4$

$\frac{1}{3} = 0.333\ldots$ or $0.\dot{3}$

$\frac{1}{12} = 0.0833\ldots$ or $0.08\dot{3}$

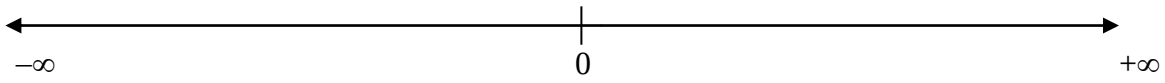
(ii) Every irrational number can only be expressed as a non-terminating and non-recurring decimal number.

E.g. $\sqrt{2} = 1.41423\ldots..$

$\pi = 3.14159\ldots..$

The Set of Real Numbers

The union of the set of Rational Numbers and the set of Irrational Numbers is called the set of Real Numbers. Its graphical (or geometrical) representation is the straight line from $-\infty$ to $+\infty$ with an origin 0.



Every real number corresponds to exactly one point on the line and every point on the line corresponds to a real number. Thus this line is called the Real Number Line and represents the **Real Number System**, denoted by R.

1.1 OPERATIONS WITH REAL NUMBERS

Let a, b, c $\in$ R, then

(i) $a + b \in R$ and $ab \in R$ – closure law .

(ii) $a + b = b + a$ – commutative law of addition +.

(iii) $(a + b) + c = a + (b + c)$ – associative law of +.

(iv) $ab = b\,a$ – commutative law of $\times$.

(v) $(ab)c = a\,(bc)$ – associative law of $\times$.

(vi) $a\,(b + c) = ab + ac$ – Distributive law of $\times$ over +.

(vii) $a + o = o + a = a$ – existence of an identity under +.

(viii) $a.1 = 1.a = 1$ – existence of an identity under $\times$.

(ix) $a + x = x + a = o$ – existence of an inverse under +

x is denoted by $(-a)$.

(x) $a.a^{-1} = a^{-1}\,a = 1$ – existence of an inverse under $\times$.

a^{-1} is denoted by $\frac{1}{a}$.

Any set, such as R, which satisfies the above rules is called a **field**

1.2 INEQUALITIES

Let a, b $\in$ R. If $a - b$ is a positive number, we say that a is greater than b or b is less than a. We write a > b or b < a.

If the possibility exists that a equals b, then we write $a \geq b$ or $b \leq a$.

Theorems on Inequalities:

If a, b, c $\in$ R, then

(i) Either $a > b$, $a = b$ or $a < b$

This is called the law of trichotomy.

(ii) If $a > b$ and $b > c$, then $a > c$

This is called the law of transitivity.

(iii) If $a > b$, then $a + c > b + c$.

(iv) If $a > b$ and $c > o$, then $ac > bc$.

(v) If $a > b$ and $c < o$, then $ac < bc$.

QUADRATIC INEQUALITIES

a) If $a, b \in R$, such that $ab > 0$, then either $a > 0$ and $b > 0$ (i.e. both are positive) or $a < 0$ and $b < 0$ (i.e. both are negative).

Example 1 Find the truth of $x^2 - x - 2 > 0$.

Solution $x^2 - x - 2 > 0$, factoring the left hand side we have

$(x+1)(x-2) > 0$

Either (i) $x + 1 > 0$ and $x - 2 > 0$

$\Rightarrow \quad x > -1$ and $x > 2$

On a number line we have

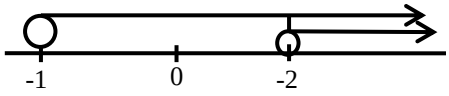


Figure 1.1

For which the truth set is $\{x: x > 2\}$ or

(ii) $x + 1 < 0$ and $x - 2 < 0$

$\Rightarrow \quad x < -1$ and $x < 2$

On a number line we have

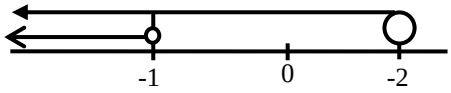


Figure 1.2

The truth set is $\{x: x > -1\}$
From (i) and (ii) the truth set is $\{x : x > 2\} \cup \{x : x < -1\}$

Method 2

Graphical method

Given $x^2 - x - 2 > 0$, factoring the left hand side, we have

$(x+1)(x-2) > 0$

Let $f(x) = (x+1)(x-2)$

For $f(x) = 0$, we have $x + 1 = 0$ or $x - 2 = 0$.

$\Rightarrow \quad x = -1$ or $x = 2$

Sketching $f(x)$, we have

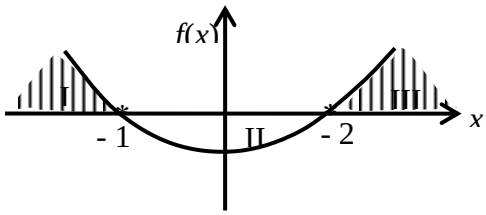


Figure 1.3

The curve has divided the entire x-axis into three regions, thus region I where $x < -1$, II where $-1 < x < 2$ and III where $x > 2$.
For the truth set of this problem we seek the region(s) where $f(x) > 0$

$$(x^2 - x - 2 > 0).$$

The regions are I and III and therefore the truth set is $\{x : x < -1\} \cup \{x : x > 2\}$.

Note:
If the question had been $(x^2 - x - 2 \geq 0)$, then the solution would have been

$$\{x : x \leq -1\} \cup \{x : x \geq 2\}.$$

Method 3

Let $f(x) = x^2 - x - 2 = (x+1)(x-2)$. $f(x) = 0$ when $x = -1$ and $x = 2$,

so consider the figures of the factors in the following intervals.

	$x < -1$	$-1 < x < 2$	$x > 2$
$x+1$	–	+	+
$x-2$	–	–	+
$f(x)$	+	–	+

Here the solution set is where $f(x) > 0$, hence the truth set is $\{x : x < -1\} \cup \{x : x > 2\}$

b) If $a, b \in R$ such that $ab < 0$, then either

(i) $a > 0$ and $b < 0$ or

(ii) $b < 0$ and $b > 0$

i.e. If the product of two numbers is negative, then the numbers are of opposite signs.

Example 2 Solve $x^2 + 4x + 3 < 0$.

Solution For $x^2 + 4x + 3 < 0$, factoring the LHS, we have

$(x+1)(x+3) < 0$

Therefore either (i) $x+1 > 0$ and $x+3 < 0$

$\Rightarrow \quad x > -1$ and $x < -3$

On a number line, we have

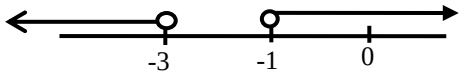


Figure 1.4

Or $x+1 < 0$ and $x+3 > 0$

$\Rightarrow \quad x < -1$ and $x > -3$

On the number line we have

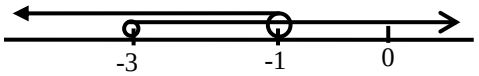


Figure 1.5

$-3 < x < -1$

Method 2: (**Graphical method**)

$x^2 + 4x + 3 < 0$, factoring the LHS, we have

$(x+1)(x+3) < 0$

Let $f(x) = (x+1)(x+3)$. Now $f(x) = 0$

$\Rightarrow \quad x + 1 = 0$ or $x + 3 = 0$

$\Rightarrow \quad x = -1$ or $x = -3$

Sketching $f(x)$, we have

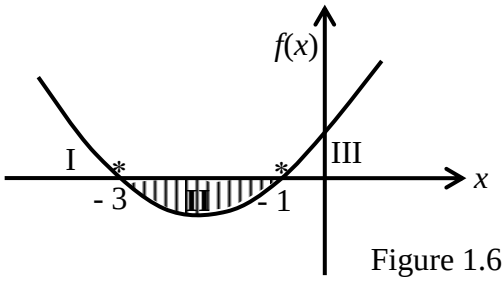


Figure 1.6

Here, for the solution we seek the region where $f(x) < 0$, which is region II, hence $-3 < x < -1$ is the solution.

Method 3: Let $f(x) = x^2 + 4x + 3 = (x+1)(x+3)$.
$f(x)$ is zero when $x = -1$ or $x = -3$,

so consider the signs of the factors in the following intervals.

	$x < -3$	$-3 < x < -1$	$x > -1$
$x+1$	–	–	+
$x+3$	–	+	+
$f(x)$	+	–	+

Here the solution set is where $f(x) < 0$ i.e. $-3 < x < -1$.

Example 3 For what values of x is $2x^2 + 5x - 3$

i) negative

ii) positive

Solution Let $f(x) = 2x^2 + 5x - 3 = (2x - 1)(x + 3)$. $f(x) = 0$ if $x = \frac{1}{2}$ or , $x = -3$,

So we consider the signs of the factors in the following intervals:

	$x < -3$	$-3 < x < \frac{1}{2}$	$x > \frac{1}{2}$
$2x-1$	–	+	+
$x+3$	–	–	+
$f(x)$	+	–	+

i) We seek the region where $f(x) < 0$ i.e. $-3 < x < \frac{1}{2}$.

ii) We seek the region where $f(x) > 0$ i.e. $x < -3 \cup x > \frac{1}{2}$

Example 4

Find the truth set of $9 - 3x - 2x^2 \leq 0$

Solution

Let $f(x) = 9 - 3x - 2x^2 = (3 - 2x)(3 + x).$

For $f(x) = 0,$ we have

$x = \frac{3}{2}$ or $x = -3$

Using the graphical method, we have

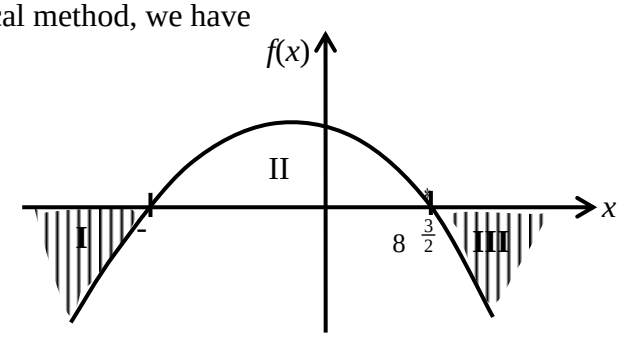


Figure 1.7

Here we seek the region(s) of the curve where $f(x) \leq 0$.
Since the coefficient of x^2 is negative, the nature of the curve is like "$\cap$".

Example 5 Solve $x^2 \geq x$.

Solution Let $f(x) = x^2 - x = x(x-1)$. For $f(x) = 0$, $x = 0$ or $x = 1$.

Using the graphical method, we have;

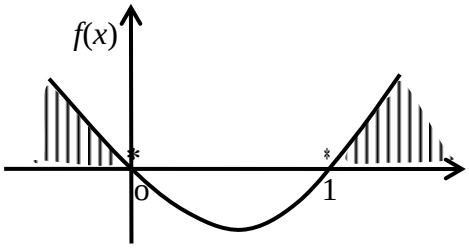


Figure 1.8

Hence, $x \leq 0 \cup x \geq 1$

Example 6 For what ranges of values of x is $2x^2 + 5x - 3$ positive?

Solution Let $y = x^2 + 5x - 3$

$= (2x-1)(x+3)$

$\therefore \; y = 0$

$\Rightarrow 2x - 1 = 0$

or $x + 3 = 0$

Thus $x = \frac{1}{2}$ or $x = -3$

i.e. the graph of the function crosses the $x-axis$ at $x = \frac{1}{2}$ and at $x = -3$

The coefficient of x^2 in the function is positive. The graph has a minimum turning point.

A sketch of the graph is as shown in Figure 1.9. From the graph, the function is positive for values of x in

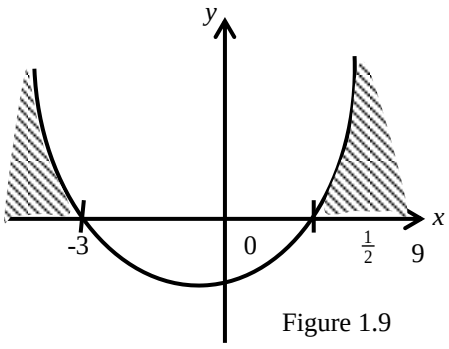


Figure 1.9

the interval, less than -3 or greater than. Thus $x < -3$ or $x > \frac{1}{2}$.

Example 1 .7 For what ranges of values of x is $(2x-3)(1-x)$ positive?

Solution Let $y=(2x-3)(1-x)$

$\therefore \quad y = 0$

$\Rightarrow \quad 2x-3 = 0$

or $\quad 1-x = 0$

$x=\frac{3}{2}$ or $x = 1$

The coefficient of x^2 in the function is negative. Therefore the graph has maximum turning point. A sketch of the graph is shown. From the graph the function is positive for values of x lying between 1 and $\frac{3}{2}$.

Thus $1 < x < \frac{3}{2}$.

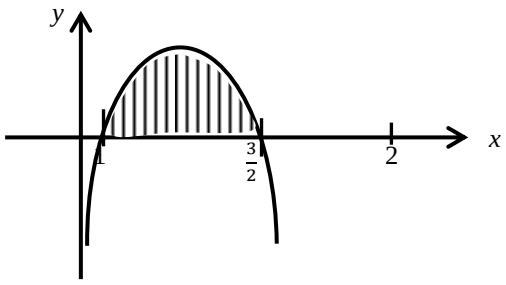


Figure 1.11

1.3 THE ABSOLUTE VALUE OF REAL NUMBERS

The absolute value of x, denoted by $|x|$ is the distance of x from zero (0). That is why absolute value is never negative. Absolute value only asks 'how far'? not in which direction.
This means that $|x| = 3$, because 3 is three units to the right of zero, also $|-3| = 3$ because – 3 is three units to the left of zero.

Simplify each of the following

(i) $|0-6| = |-6| = 6$

(ii) $|5-2| = |3| = 3$

(iii) $|2-5| = |-3| = 3$

(iv) $|0(-4)| = |0| = 0$

(v) $|2+3(-4)| = |-10| = 10$

(vi) $-|-4| = -(4) = -4$

(vii) $-|(-3)^2| = -|9| = -9$

Let $x \in R$. The absolute value of x, denoted by $|x|$, is defined as

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} . \qquad (1)$$

The following theorems hold for inequalities.

For $x, y \in R$,

(i) $|xy| = |x|\,|y|$

(ii) $|x + y| \leq |x| + |y|$, called the triangle inequality.

(iii) $|x - y| \geq |x| - |y|$

(iv) $|x| = 0 \Leftrightarrow x = 0$.

Note: The property

(i) may be extended to n real numbers, namely

$$|x_1 . x_2 \ \ldots \ x_n| = |x_1|\,|x_2| \ \ldots \ |x_n| ;$$

(ii) may be extended to n real numbers as

$$|x_1 + x_2 + \ \ldots \ + x_n| \leq |x_1| + |x_2| + \ \ldots \ + |x_n|$$

(iii) For the case, $|x - y|$ may be geometrically interpreted as the distance between any two points x and y on the number line. It is clear then that

$$|x - y| = |y - x| .$$

Worked Examples

1. Prove that (i) $|xy| = |x||y|$

(ii) $|x + y| \leq |x| + |y|$

Solution

(i) We consider the four cases:

$x = 0, \quad y = 0$

$x > 0, \ y > 0$

$x < 0, \ y > 0$

$x < 0, \ y < 0.$

If $x = 0$ and $y = 0$, $xy = 0.0 = 0$

$\therefore |xy| = |0| = |0||0| = |x|\,|y|.$

If $x > 0$ and $y > 0$, $xy > 0$, $|x| = x$, $|y| = y$ and $|xy| = xy = |x|\,|y|.$

If $x < 0$ and $y > 0$, then $|x| = -x$ and $|y| = y$ and $xy < 0$

$\Rightarrow |xy| = -(xy) = (-x)(y) = |x|\,|y|$.

If $x < 0$ and $y < 0$, then $|x| = -x$ and $|y| = -y$ and $xy > 0 \Rightarrow |xy| = xy$

But $|x|\,|y| = (-x)(-y) = xy$

$\therefore |xy| = |x|\,|y|$.

This completes the proof.

(ii) We again consider the cases

$x = 0$, $y = 0$

$x > 0$, $y > 0$

$x < 0$, $y > 0$

$x < 0$, $y < 0$

If $x = 0$ and $y = 0$, then $|x + y| = |0 + 0| = |0| = 0$.

Also $|x| = |0| = 0$ and $|y| = |0| = 0$

$\therefore |x| + |y| = 0 + 0 = 0$

$\therefore |x + y| = |x| + |y|$.

If $x > 0$ and $y > 0$, then $x + y > 0 \Rightarrow |x + y| = x + y$.

Also $|x| = x$ and $|y| = y$, so $|x| + |y| = x + y$.

$\therefore |x + y| = |x| + |y|$.

If $x < 0$ and $y > 0$; and suppose that $|x| > |y|$,

then $x + y < 0 \Rightarrow |x + y| = -(x + y) = -x + (-y)$.

Also $|x| = -x$ and $|y| = y \Rightarrow |x| + |y| = -x + y$

Clearly, $|x + y| - (|x| + |y|) = -2y < 0$

$\therefore |x + y| < |x| + |y|$

On the other hand if $|x| < |y|$, then

$x + y > 0 \Rightarrow |x + y| = x + y$

Also $|x| = -x$ and $|y| = y \Rightarrow |x| + |y| = -x + y$.

$\therefore |x + y| - (|x| + |y|) = 2x < 0$

$\Rightarrow |x + y| < |x| + |y|$

Finally, if $|x| = |y|$, then $x + y = 0$.

So that $|x + y| = |0| = 0$.

But

$$|x| + |y| = -x + y > 0$$

$$\therefore |x + y| < |x| + |y|$$

Now, if $x < 0$ and $y < 0$, then $x + y < 0$

$$\Rightarrow |x + y| = -(x + y) = (-x) + (-y) = |x| + |y|$$

Thus for all $x, y \in \Re$,

$$|x + y| \leq |x| + |y|$$

This completes the proof.

EXERCISE 1

1. Find the truth set of $x^2 - 7x + 10 \geq 0$.
2. Solve $6x^2 - x - 35 \leq 0$.
3. Solve $(2x - 1)(x + 2) < 0$.
4. Solve $(1 - 2x)(x + 2) < 0$.
5. Solve $2x^2 + x - 6 \leq 0$.
6. Find the truth $35 - 2x - x^2 \geq 0$.

   Note: (We can multiply both sides of this inequality by – 1)

   i.e. $x^2 + 2x - 35 \leq 0$ and find the truth set of $x^2 + 2x - 35 \leq 0$, which is the same as $35 - 2x - x^2 \geq 0)$.
7. Find the truth set of $3x^2 - x + 2 > 0$.
8. Find the truth set of $(3 - 2x)(4 - 2x) \leq 0$.

9. Prove that $\sqrt{2}$ is irrational.
10. Prove that $\sqrt{3}$ is irrational.

CHAPTER 2

POINT SETS

2.1 Definition

Any set whose members are real numbers is called a **one-dimensional point set**. All the elements of a point set can be located on the real number line.

E.g. The sets {0, 1, 2, 3, 4, 5}, {x: $-1 < x < 0$} are point sets.

2.2 Intervals

The set of values of x such that $a \le x \le b$, where $a, b \in R$, is called a **closed interval**; it is denoted by $[a, b]$. The set $\{x: a < x < b, a, b \in R\}$ is called an **open interval** and is denoted by (a, b). The half-open or half closed intervals are $\{x: a < x \le b\} \equiv (a, b]$.

$\{x: a \le x < b\} \equiv [a, b)$.

The intervals indicated above are said to be bounded.

There are unbounded intervals such as:

$\{x : -\infty < x < b\} \equiv (-\infty, b)$ or $\{x: x < b\}$,

$\{x: a < x < \infty\} \equiv \{x: x > a\} \equiv (a, \infty)$

$\{x: -\infty < x < \infty\} \equiv \{x: x \in R\} \equiv R$, or $(-\infty, \infty)$.

2.3 NEIGHBOURHOODS

The set of all points x such that $|x - a| < \delta$, where $a \in R$ and $\delta > 0$ is called a **δ–neighbourhood of the point *a***

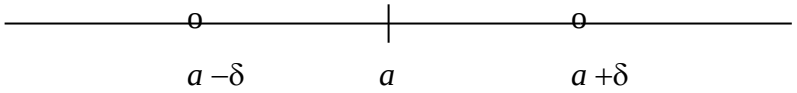
$|x - a| < \delta$

$\Rightarrow -\delta < x - a < \delta$

$\Rightarrow a - \delta < x < a + \delta$

Note

The δ–neighbourhood of a is an interval with centre a and length 2δ .



δ–Neighbourhood of a

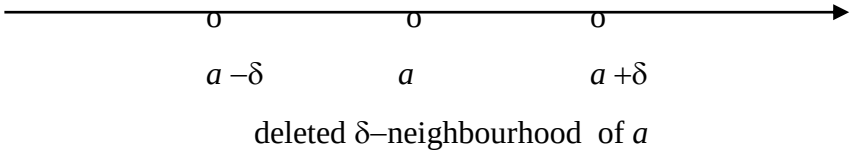
(ii) The set of all points x such that

$$0 < |x - a| < \delta$$

is called a **deleted δ–neighbourhood of *a*** .

Note

The deleted δ–neighbourhood of a is an interval with centre a and length 2δ , with the point *a* excluded from the interval.



deleted δ–neighbourhood of *a*

Examples:

(a) The 1 – neighbourhood of 0 is the open interval (–1, 1). It has length 2 and centre 0.

(b) The deleted 2 – neighbourhood of 0 is the interval $\{x: -2 < x < 2, x \neq 0\}$.

2.4 LIMIT POINT, POINT OF ACCUMULATION OR CLUSTER POINT

Definition

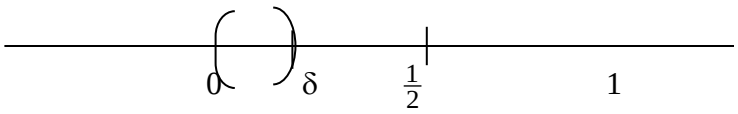
A *limit point*, *point of accumulation*, or *cluster point* of a set A of numbers is a number *l* such that every deleted δ–neighbourhood of *l* contains elements of the set A. That is, for every $\delta > 0$, $\exists\ x \in A$ s.t. $0 < |x - l| < \delta$. Since δ may be as small as we please, it is clear that every δ–neighbourhood of 0 contains an infinity of elements of A.

Examples

1) (a) 0 is a limit point of the set

A = (0, 1),

Since every deleted δ–neighbourhood of 0 has elements of A.



It is clear that $\{x : -\delta < x < \delta$ will always have elements of A, $x \neq 0\}$

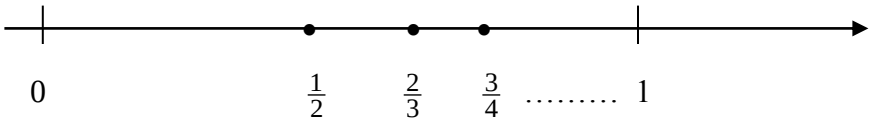
(b) 1 is a limit point of the set

A = (0, 1), since every deleted δ–neighbourhood of 1 has elements of A.

(c) All the numbers in the closed interval **[0, 1]** are cluster/ limit points of the open interval **(0, 1).**

2) 1 is a limit point of the set

$$B = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots\right\}$$



Clearly, for every deleted δ–neighbourhood of 1, there will always be elements of the set B.

Note

(i) No finite set has a limit point

E.g. The set {1, 2, 3, 4, 5} has no limit points since for any suggested number, a deleted δ–neighbourhood can be constructed with no elements of the set present.

(ii) An infinite set may or may not have a limit point. In particular, the natural numbers N = {1, 2, 3, … } has no cluster points. The set of rational numbers has infinitely many limit points.

(iii) If a set contains all its limit points, it is called a **closed set.**

E.g. the set [0, 1] is a closed set, while the set (0, 1) is **not** a closed set since 0 and 1 even though are limit points do not belong to the set.

Exercises:

(1) How many limit points does the set (–1, 1) have? Why is 0 a limit point?

(2) Does the set

$A = \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$ have any limit points? If yes, find them.

(3) Why does the set

$\{-10, -9, -8, \ldots, 0, 1, 2, 3, \ldots, 10\}$ not have a limit point. Show why the point $x = \frac{1}{2}$ cannot be a limit point of the set.

(4) An infinite set is defined by

$$A = \{ \frac{1}{2^n} : n = 0, 1, 2, 3, \ldots \}.$$

Does it have a limit point? If yes, determine it. Is the set A closed? Why?

2.5 BOUNDS

Let A be a set. If $\exists$ M (real) such that $x \leq M$, $\forall x \in A$, then A is said to be **bounded above**. M is called an **upper bound of the set A**. If $\exists$ m (real) such that $x \geq m$, $\forall x \in A$, then A is said to be **bounded below**. m is called a **lower bound** of the set A.

Let M be an upper bound of A. Then if for any $\in > 0$ (however small), $\exists$ at least one element x in A such that $x > \underline{M} - \in$, the number $\underline{M}$ is called the **least upper bound** (l.u.b.) of A.

Let $\overline{m}$ be a lower bound of A, then if for any $\in > 0$ (however small), $\exists$ at least one element x in A such that $x < \overline{m} + \in$, the number $\overline{m}$ is called the **greatest lower bound** (g.l.b.).
If $\forall x \in A$, $\exists m$, M (real) such that $m \leq x \leq M$, then the set A is said to be **bounded**.

Examples

(1) The set [0, 1] is a bounded set since for example, $-2 < x < 3$, $\forall x \in [0, 1]$.

The least upper bound is 1

The greatest lower bound is 0.

(2) The set

$$A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots \right\}$$

is a bounded set, since $-1 < x < 2$, $\forall x \in A$. The least upper bound is 1 and the greatest lower bound is 0.

2.6 THE BOLZANO -WEIERSTRASS THEOREM

The Bolzano -Weierstrass theorem states that every bounded infinite set has at least one limit point.

Examples

(1) Show that $S = \{\frac{1}{n} : n = 1, 2, 3, \ldots\}$ satisfies the Weierstrass – Bolzano theorem. Determine the limit and prove it.

Solution

$$S = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}.$$

S is bounded above by 2 and bounded below by –1. $\therefore$ S is bounded.

Also S is infinite set since it can be placed in 1 – 1 correspondence with a subset of itself.

$\therefore$ S has at least one limit point, by the Bolzano – Weierstrass Theorem.

The limit point is 0.

A deleted δ–neighbourhood of 0 is

$$0 < \left|\frac{1}{n} - 0\right| < \delta, \quad n \in N.$$

$$\Rightarrow 0 < \frac{1}{n} < \delta \Rightarrow n\delta > 1$$

$$\Rightarrow n > \frac{1}{\delta}$$

It is clear that for every δ, there will always be an integer $N(\delta)$

$$\left|\frac{1}{n} - 0\right| < \delta \quad , \quad \forall\, n > N(\delta).$$

(2) Does the set $T = \{\frac{1}{2^n} : n = 1, 2, 3, \ldots\}$ satisfy the Bolzano – Weierstrass theorem? Why? Does the set T have a limit point? If yes, find it and prove it.

Solution

$$T = \left\{ \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}, \quad \ldots \right\}$$

Clearly T is bounded. For example, $-1 < x < 1, \ \forall\ x \in T$. T is also infinite and so satisfies the Weierstrass-Bolzano theorem. Therefore, T has at least one limit point. In fact, T has only one limit point 0.

To prove that 0 is a limit point of T, we show that every deleted δ-neighbourhood of 0 has elements of T.

$$0 < \left| \frac{1}{2^n} - 0 \right| < \delta, \ n \in N$$

$$\frac{1}{2^n} < \delta$$

$$2^n > \frac{1}{\delta}$$

$$\Rightarrow \quad n \log 2 > \log\left(\frac{1}{\delta}\right)$$

$$n > \frac{\log\left(\frac{1}{\delta}\right)}{\log 2}$$

It is clear that for every δ, there will always be an integer $N(\delta)$ such that $\left|\frac{1}{2^n} - 0\right| < \delta$.

The existence of an integer N(δ) immediately greater than $\frac{\log\left(\frac{1}{\delta}\right)}{\log 2}$ completes the proof.

EXERCISE 2

1. Define a deleted $\frac{1}{2}$ – neighbourhood of 2.

2. How many cluster points does the interval $(-1, 2)$ have?

3. Given that $A = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\ldots\right\}$, find the

(i) lower bound of A.
(ii) greatest lower bound of A.
(iii) upper bound of A.
(iv) least upper bound of A.

4. Find the cluster point(s) of $1, \frac{1}{2}, 1, \frac{1}{3}, 1, \frac{1}{4}, 1, \frac{1}{5}\ldots$

5. If $T = \left\{\frac{1}{5^n} : n = 0, 1, 2, 3\ldots\right\}$, then T is closed?

A. True
B. False

6. Find the limit point(s) of $a_n = 1 + (-1)^n + \frac{1}{n}$

7. The set $T = \{1, 2, 3, 4, 5\}$ has no cluster point.

A. True
B. False

8. Find the cluster point(s) of

(i) $A = \left\{1 + (-1)^n \left(\frac{n^2 + 2}{2n^2 + 1}\right); \quad n = 1, 2...\right\}$

(ii) $B = \left\{(-1)^n \left(1 + \frac{1}{n}\right); \quad n = 1, 2...\right\}$

(iii) $C = \left\{1 + (-1)^n; \quad n = 1, 2...\right\}$

9. (i) State Bolzano-Weierstrass theorem.

(ii) Does the set $T = \left\{\frac{1}{3^n} : n = 1, 2, 3\ldots\right\}$ satisfy that Bolzano-Weierstrass theorem?

Does the set T have a cluster point? If yes, find it and prove it.

CHAPTER 3

SEQUENCES

3.1 DEFINITION

A set of numbers $U_1, U_2, U_3, \ldots$ in a definite order of arrangement and formed by a definite rule is called a **sequence of numbers.**

Every number in the sequence is called a term of the sequence. If the sequence has a last term, then it is called a **finite sequence**. If the number of terms is infinite, then the sequence is called an **infinite sequence**. The nth term of the sequence will be denoted by U_n, while the sequence itself will be denoted by $\{U_n\}$.

Examples

(i)

$\{1, 2, 3, \ldots, 20\}$ is a finite sequence of the counting numbers up to 20.

(ii) $\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\}$ is an infinite sequence with the nth term $U_n = \frac{1}{n}$.

3.2 TYPES OF SEQUENCES

(a) Arithmetic Sequence

A sequence $\{U_n\}$ of numbers is said to be an arithmetic sequence if

$$U_{n+1} - U_n = d \quad , \forall\, n \qquad \text{----------} \qquad (1)$$

where d is a constant. The constant d is called the **common difference.**

The sum of the first *n* terms of an Arithmetic sequence is given by

$$S_n = \frac{n}{2}\{2a + (n-1)d\}, \qquad \text{----------} \qquad (2)$$

where a is the first term of the sequence.

E.g.

The sequence

2, 5, 8, 11, …

is an arithmetic sequence with $a = 2$ and $d = 3$.

The sum of the first 10 terms is:

$S_{10} = \frac{10}{2} \{2 (2) + (10 - 1) (3) \}$

$= 5 \{4 + 27\}$

$= 5 \{31\}$

$= \underline{\underline{155}}$

(b) Geometric Sequence

A sequence $\{U_n\}$ is said to be a geometric sequence if

$$\frac{U_{n+1}}{U_n} = r, \qquad \forall n \qquad \text{----------(3)}$$

where r is a constant. The number r is called the common ratio.

The sum of the first n terms of a geometric sequence is given by

$$S_n = \frac{a(1-r^n)}{1-r}, \quad \text{for } r < 1 \qquad \text{---------- (4a)}$$

and $$S_n = \frac{a(r^n - 1)}{r-1}, \quad \textit{for } r > 1 \qquad \text{---------- (4b)}$$

where a is the first term.

E.g. The sequence

$$1, \ \frac{1}{2}, \ \frac{1}{4}, \ \frac{1}{8}, \ \ldots$$

is a geometric sequence with $a = 1$ and $r = \frac{1}{2}$.

SUM TO INFINITY

The sum to infinity of a geometric sequence dented by

$$S_\infty = \frac{a}{1-r} \qquad \text{---------- (4c)}$$

(c) Sequences which are neither arithmetic nor geometric

There are sequences which are neither arithmetic nor geometric. In what follows general properties of sequences will be discussed.

3.3 LIMIT OF A SEQUENCE

An infinite sequence $\{U_n\}_{n=1}^{\infty}$ is said to have a limit l as n approaches ∞ if U_n can be made as close to l as we please by choosing n sufficiently large.

We then write.

$$\lim_{n\to\infty} U_n = l \,. \qquad \text{----------} \qquad (5)$$

The above definition may be more rigorously written as follows: $\lim_{n\to\infty} U_n = l$ if for every $\in > 0$ (however small), $\exists$ a positive integer N (depending on $\in$) s.t. $|U_n - l| < \in$, $\forall$ n > N.

Examples

(1) The sequence $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ has limit 0. Thus $U_n = \frac{1}{n}$ and $l = 0$.

The difference between U_n and l is $\left|\frac{1}{n} - 0\right| < \varepsilon$

$$\left|\frac{1}{n}\right| < \varepsilon$$

$$\frac{1}{n} < \varepsilon \,, \Rightarrow n > \frac{1}{\varepsilon}$$

Thus any choice of n greater than $\frac{1}{\varepsilon}$ would satisfy the required condition.

It is clear that no matter how small the difference $\left|\frac{1}{n} - 0\right|$ is made there will always be the number N.

(2) The sequence $\left\{\frac{n+1}{n}\right\}$ has limit 1.

(3) The sequence $\left\{\frac{n^2+1}{n}\right\}$ has no limit.

Note:

If a sequence has a limit then it is called a **convergent sequence**; otherwise it is called a **divergent sequence**.

3.4 THEOREMS ON SEQUENCES

Theorem 1

If a sequence $\{U_n\}$ has a limit, then the limit is unique.

(In other words, a convergent sequence has only one limit).

Proof:

Suppose that $\{U_n\}$ has two limits l_1 and l_2. Then $\forall \in > 0$, $\exists$ N (a + ve integer) such that

$|\ U_n - l_1\ | < \varepsilon, \quad \forall\ n > N$, and

$|\ U_n - l_2\ | < \varepsilon\ , \ \forall\ n > N.$

Then

$$|\ l_1 - l_2\ | = |\ l_1 - U_n\ + U_n - l_2\ |$$
$$\leq |\ l_1 - U_n\ | + |\ U_n - l_2\ |$$
$$< \quad \varepsilon \quad + \quad \varepsilon$$

i.e. $|\ l_1 - l_2\ | < 2\ \varepsilon$

Since $\in > 0$ (however small), it follows that $|\ l_1 - l_2\ |$ is less than every positive integer.

$\therefore\ |\ l_1 - l_2\ | = 0 \ \Rightarrow\ l_1 - l_2 = 0 \ \Rightarrow\ l_1 = l_2$.

This completes the proof.

Theorem 2

Let $\{U_n\}$ and $\{V_n\}$ be two sequences such that $\lim_{n\to\infty} U_n = A$ and $\lim_{n\to\infty} V_n = B$, Then the following results hold:

(1) $$\lim_{n\to\infty}(U_n + V_n) = \lim_{n\to\infty} U_n + \lim_{n\to\infty} V_n$$
$$= A + B.$$

(2) $$\lim_{n\to\infty}(U_n - V_n) = \lim_{n\to\infty} U_n - \lim_{n\to\infty} V_n$$
$$= A - B$$

(3) $$\lim_{n\to\infty}(U_n \times V_n) = \lim_{n\to\infty} U_n \times \lim_{n\to\infty} V_n$$
$$= A \times B$$

(4) $$\lim_{n\to\infty}\left(\frac{U_n}{V_n}\right) = \frac{\lim_{n\to\infty} U_n}{\lim_{n\to\infty} V_n} = \frac{A}{B}$$ provided that $\lim_{n\to\infty} V_n = B \neq 0$.

(a) If B = 0 and A ≠ 0, then $\lim_{n\to\infty} \frac{U_n}{V_n}$ does not exist.

(b) If B = 0 and A = 0, then $\lim_{n\to\infty} \frac{U_n}{V_n}$ may or may not exist.

(5) $$\lim_{n\to\infty} U_n^p = \left(\lim_{n\to\infty} U_n\right)^p = A^p$$

(6) $$\lim_{n\to\infty} p^{u_n} = p^{\lim_{n\to\infty} u_n} = p^A.$$

For the conditions (5) and (6), p must be real and A^p or p^A must exist.

3.5 BOUNDED, MONOTONIC SEQUENCES

A sequence $\{U_n\}$ is said to be bounded above if $\exists$ M (a constant) such that $U_n \leq M$, $\forall n$. M is called an upper bound. If $\exists$ m a constant such that $U_n \geq m$, $\forall$ n , then the sequence $\{U_n\}$ is said to be bounded below and m is called a lower bound. If $\exists$ m, M such that $m \leq U_n \leq M$, $\forall n$, then the sequence $\{U_n\}$is said to be a <u>bounded sequence</u>.

An upper bound <u>M</u> is called the least upper bound (l. u. b.) of the sequence if for every $\varepsilon > 0$, there will always exist at least one term of the sequence greater than $M - \varepsilon$.

A lower bound $\overline{m}$ is called the greatest lower bound (g. l. b.) of the sequence if for every $\varepsilon > 0$, $\exists$ at least one term of the sequence is less than $\overline{m} + \varepsilon$.

Example

The sequence $\left\{\frac{2n+1}{n}\right\} \equiv \left\{2 + \frac{1}{n}\right\}$ is bounded below by 1 and bounded above by 4. Thus $\left\{\frac{2n+1}{n}\right\}$ is a bounded sequence.

The least upper bound is 3 and the greatest lower bound is 2.

A sequence $\{U_n\}$ is said to be a **monotonic increasing sequence or a non-decreasing sequence** if $Un \leq U_{n+1}$, $\forall n$. If $U_{n+1} \leq U_n$, $\forall$ n, then the sequence is called a **monotonic decreasing sequence or a non-increasing sequence**.

If $U_n < Un_{+1,}$ $\forall n$, the sequence is said to be strictly increasing. If $Un_{+1} < Un_{,}$ $\forall n$, the sequence is strictly decreasing.

If a sequence is either monotonic increasing or monotonic decreasing, it is said to be a **monotonic sequence.**

Example

Show that the sequence $\left\{\frac{n+1}{n}\right\}$ is:

(i) a monotonic decreasing sequence

(ii) a bounded sequence.

Solution

(i) $U_n = \frac{n+1}{n}$

$$U_{n+1} - U_n = \frac{n+2}{n+1} - \frac{n+1}{n}$$

$$= \frac{n(n+2) - (n+1)(n+1)}{n(n+1)}$$

$$= \frac{n^2 + 2n - n^2 - 2n - 1}{n(n+1)}$$

$$= \frac{-1}{n(n+1)} \leq 0$$

$\Rightarrow U_{n+1} - U_n \leq 0$

$\Rightarrow U_{n+1} \leq U_n, \quad \forall n.$

$\therefore$ $\{U_n\}$ is a monotonic decreasing sequence.

(ii) $\left\{\frac{n+1}{n}\right\} \equiv \left\{\frac{2}{1}, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \cdots\right\}$

The sequence is bounded above by 3 (say) and bounded below by 0 (say).

$\therefore$ The sequence is bounded.

Theorem: Every bounded monotonic sequence has a limit.

Example

Use the above theorem to establish that the sequence $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$ has a limit.

Prove that $\lim_{n\to\infty}\left(\frac{n}{n+1}\right) = 1$.

Solution

The sequence $\left\{\frac{n}{n+1}\right\} \equiv \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots\right\}$ is bounded above by 2(say) and bounded below by 0 (say). $\therefore$ the sequence is bounded. The sequence is monotonic increasing since

$$U_{n+1} - U_n = \frac{n+1}{n+2} - \frac{n}{n+1}$$

$$= \frac{(n+1)^2 - n(n+2)}{(n+1)(n+2)}$$

$$= \frac{1}{(n+1)(n+2)} \geq 0, \ \forall \text{ n}$$

$\Rightarrow \quad U_{n+1} - U_n \geq 0$

$\Rightarrow U_{n+1} \geq U_n, \quad \forall$ n.

Since the sequence is bounded and monotonic, it has a limit.

To prove that $\lim_{n\to\infty}\left(\frac{n}{n+1}\right) = 1$, we wish to show that for every $\delta > 0$, $\exists$ a positive integer N, such that $\left|\frac{n}{n+1} - 1\right| < \delta \qquad \forall$ n > N.

Now $\left|\frac{n}{n+1} - 1\right| = \left|\frac{n-(n+1)}{n+1}\right| = \left|\frac{-1}{n+1}\right| = \frac{1}{n+1}$.

For any $\delta > 0$

$\frac{1}{n+1} < \delta$

$\Rightarrow (n + 1)\,\delta > 1$

$\Rightarrow n > \frac{1}{\delta} - 1$.

Thus choosing N a positive integer immediately larger than $(\frac{1}{\delta} - 1)$, the result follows.

Example

Prove that

(a) $\lim_{n\to\infty}\left(\frac{4-2n}{3n+2}\right) = -\frac{2}{3}$ (b) $\lim_{n\to\infty}\left(\frac{3}{4}\right)^n = 0$

Solution

(a) For every $\delta > 0$

$$\left|\frac{4-2n}{3n+2} - \left(\frac{-2}{3}\right)\right| < \delta$$

$$\Leftrightarrow \left|\frac{4-2n}{3n+2} + \frac{2}{3}\right| < \delta$$

$$\Leftrightarrow \left|\frac{3(4-2n) + 2(3n+2)}{3(3n+2)}\right| < \delta$$

$$\Leftrightarrow \left|\frac{16}{3(3n+2)}\right| < \delta \Leftrightarrow \frac{16}{3(3n+2)} < \delta$$

$$\Leftrightarrow (3n + 2)\,\delta > \frac{16}{3} \Leftrightarrow 3n + 2 > \frac{16}{3\delta}$$

$$\therefore n > \frac{1}{3}\left(\frac{16}{3\delta} - 2\right).$$

Hence choosing N immediately larger than $\frac{1}{3}\left(\frac{16}{3\delta} - 2\right)$, the result follows.

(b) For every $\delta > 0$,

$$\left| \left(\frac{3}{4}\right)^n - 0 \right| < \delta$$

$$\Rightarrow \quad \left(\frac{3}{4}\right)^n < \delta$$

$$\Rightarrow \quad n \log_e \left(\frac{3}{4}\right) < \log_e \delta$$

$$\Rightarrow \quad n > \frac{\log_e \delta}{\log_e \frac{3}{4}} \quad (\text{since } \log_e \frac{3}{4} < 0) .$$

Hence choosing N, a positive integer immediately larger than $\frac{\log_e \delta}{\log_e \frac{3}{4}}$, the result follows.

3.6 EVALUATING LIMITS OF SEQUENCES

The limit l of a sequence $\{U_n\}$, if it exists, may be evaluated using the theorems on limits. Consider the following examples.

Examples
Evaluate the following:

(a) $\lim_{n\to\infty} \left\{ \frac{4 - 2n - 3n^2}{4n^2 + 5n} \right\}$ (b) $\lim_{n\to\infty} \left\{ \sqrt{\frac{n^2(n-1)}{1 + \frac{1}{2}n^3}} \right\}$

(c) $\lim_{n\to\infty} \left\{ \sqrt{n^2 + n} - n \right\}$ (d) $\lim_{n\to\infty} \left\{ 2^{\frac{n}{n^2+1}} \right\}$.

Solutions

(a) Using the theorems on limits we have

$$\lim_{n\to\infty} \frac{4 - 2n - 3n^2}{4n^2 + 5n} = \lim_{n\to\infty} \frac{\frac{4}{n^2} - \frac{2}{n} - 3}{4 + \frac{5}{n}}$$

$$= \frac{\lim_{n\to\infty}\left(\frac{4}{n^2} - \frac{2}{n} - 3\right)}{\lim_{n\to\infty}\left(4 + \frac{5}{n}\right)} = \frac{3}{4}.$$

(b) $$\lim_{n\to\infty} \sqrt{\frac{n^2(n-1)}{1 + \frac{1}{2}n^3}} = \sqrt{\lim_{n\to\infty}\left(\frac{n^2(n-1)}{1 + \frac{1}{2}n^3}\right)}$$

$$= \sqrt{\lim_{n\to\infty}\left(\frac{1 - \frac{1}{n}}{\frac{1}{n^3} + \frac{1}{2}}\right)} = \sqrt{\frac{1}{\frac{1}{2}}}$$

$$= \underline{\underline{\sqrt{2}}}$$

(c) $$\lim_{n\to\infty}\left\{\sqrt{n^2+n} - n\right\} = \lim_{n\to\infty} \frac{\left(\sqrt{n^2+n} - n\right)\left(\sqrt{n^2+n} + n\right)}{\sqrt{n^2+n} + n}$$

$$= \lim_{n\to\infty}\left(\frac{\left(n^2+n\right) - n^2}{\sqrt{n^2+n} + n}\right)$$

$$= \lim_{n\to\infty} \frac{n}{\sqrt{n^2+n} + n}$$

$$= \lim_{n\to\infty} \frac{1}{\sqrt{1+\frac{1}{n}} + 1} = \frac{1}{2}$$

(d) $$\lim_{n\to\infty}\left(2^{\frac{n}{n^2+1}}\right) = 2^{\lim_{n\to\infty}\left(\frac{n}{n^2+1}\right)}$$

$$= 2^{\circ}$$

$$= 1$$

EXERCISE 3

(1) Let $U_n = \dfrac{n+3}{2n+1}$

(a) Prove that the sequence $\{U_n\}$ is

(i) bounded

(ii) monotonic decreasing.

(b) Guess the limit of $\{U_n\}$ and prove it.

(2) Prove that $\lim_{n\to\infty}\left(\dfrac{n+1}{3n-2}\right) = \dfrac{1}{3}$, using the definition of a limit.

(3) Prove that $\lim_{n\to\infty} 2^{\frac{1}{n}} = 1$, using the definition of a limit.

(4) Evaluate the following:

(a) $\lim_{n\to\infty}\left(\dfrac{5n^2+3n+3}{2-n+7n^2}\right)$ (b) $\lim_{n\to\infty}\left(\dfrac{1}{n^2}\left(\dfrac{n^2-1}{n^2}\right)\right)$

(c) $\lim_{n\to\infty}\left(\dfrac{\sqrt{3n^2-4n+5}}{n+1}\right)$ (d) $\lim_{n\to\infty}\left(5^{\sqrt{\left(1+\frac{1}{n^2}\right)}}\right)$

CHAPTER FOUR

FUNCTIONS

4.1 Definition

A real function f is a relation which takes elements from a point set D (called the domain) to elements of another point set (called the co-domain) such that each element in the domain has exactly one image in the co-domain.

Any number $x \in D$ is called an independent variable. The function may be represented in the form

$$f : x \to f(x) \quad \text{or} \quad f : x \to y$$

The domain of the function is also defined as the set of all real number for which the function f is defined. When we write y = f(x) to represent a function, then y is called the dependent variable.

4.2 The Graph of a function

The graph of the function y = f(x) is a pictorial representation of the function. This representation may be on a rectangular co-ordinate system, the points being defined by the pairs (x, y) or (x, f(x)).

4.3 BOUNDED FUNCTIONS

Let $x \in [a, b]$. If $\exists$ M (real) such that $f(x) \leq M$, $\forall x \in [a, b]$, then f(x) is said to be bounded above in [*a*, b]. M is called an upper bound of f.

If $\exists$ m (real) such that $f(x) \geq m$, $\forall x \in [a, b]$ we say that f(x) is bounded below in (*a*, b) and m is called a lower bound of f.

If $\exists$ m, M such that $m \leq f(x) \leq M$, $\forall x \in [a, b]$ then the function f is said to be bounded.

If f(x) has an upper bound, it has a least upper bound; if it has a lower bound it has a greatest lower bound.

Examples

Consider the interval [–1, 1].

1. The function f(x) = 4 – x is bounded above by 5 (or any number greater than 5) and bounded below by 3 (or any number less than 3).

4.4 MONOTONIC FUNCTIONS

A function f(x) is called a monotonic increasing function in an interval [*a*, b] if for $\forall x_1, x_2 \in$ [a, b] such that $x_1 < x_2$, $f(x_1) \leq f(x_2)$.

f(x) is called monotonic decreasing if for $x_1 < x_2$, $f(x_1) \geq f(x_2)$.

If $x_1 < x_2$ and $f(x_1) < f(x_2)$, then f(x) is strictly increasing; if $f(x_1) > f(x_2)$, then f(x) is strictly decreasing.

Examples

(1) The function $f(x) = 2x + 1$ is monotonic increasing in the interval [0, 1].

(2) The function $f(x) = \frac{1}{x}$ is monotonic decreasing in the interval [1, 3].

(3) The function

$$f(x) = \frac{x + 1}{x}$$

is monotonic decreasing in the interval [1, 5].

MAXIMA AND MINIMA

Let $x_0 \in [a, b]$. If a function f(x) is such that $f(x) \leq f(x_0)$, $\forall\ x \in [a, b]$, then f(x) is said to have an **absolute maximum** in the interval $[a, b]$ at $x = x_0$. The maximum value is $f(x_0)$. If $f(x) \geq f(x_0)$, $\forall\ x \in [a, b]$, then f(x) is said to have **an absolute minimum** in the interval $[a, b]$ at $x = x_0$. The minimum value is $f(x_0)$.

If the above statements are true only for values of x in some deleted δ - neighbourhood of x_0 then f(x) is said to have a **relative maximum** or a **relative minimum** at x_0.

4.6 TYPES OF FUNCTIONS

There are three major types of functions: Polynomial functions, algebraic functions and transcendental functions.

1. **Polynomial Functions**

These are functions of the form

$P_n(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n,\ a_n \neq 0.$ ---------- (1)

This is called a polynomial in x of degree n. The numbers $a_0, a_1, \ldots, a_n$ are constants and n is a positive integer.

Theorem

Every polynomial equation $f(x) = 0$ has at least one root.

If the degree of the polynomial is n, then the equation $P_n(x) = 0$ has exactly n roots.

2. **Algebraic Function**

Any function $y = f(x)$ satisfying an equation of the form

$P_0(x)\, y^n + P_1(x)\, y^{n-1} + \ldots + P_{n-1}(x) y + P_n(x) = 0$, ---------- (2)

where $P_0(x), \ldots, P_n(x)$ are polynomials in x, is called an algebraic function.

If an algebraic function may be expressed as the quotient of two polynomials,

i.e. $P(x)/Q(x)$, where $P(x)$ and $Q(x)$ are polynomials, then the function is called a rational algebraic function. Otherwise it is an irrational algebraic function.

3. Transcendental Functions

Any function which is not algebraic is called a Transcendental Function. A transcendental function is expressible as an infinite sum of polynomials.

Examples

(a) $2x^4 + x^3 - 2x^2 + x + 9$ ---------- a polynomial of degree 4.

(b) $y = \frac{1}{x^3}$ is an algebraic function since

$x^3 \cdot \frac{1}{x^3} - 1 = 0$ (x^3, -1 are polynomials)

(c) $y = \frac{x}{x+1}$ is algebraic, since

$x \cdot \frac{x}{x+1} + \frac{x}{x+1} - x = 0$

$\Rightarrow (x+1)\frac{x}{x+1} - x = 0$

4.7 ELEMENTARY TRANSCENDENTAL FUNCTIONS

1. **Exponential Functions**.

These are functions of the form

$f(x) = a^x$, $a \neq 0, 1$

2. **Logarithmic Functions**

These have the form

$f(x) = \log_a x$, $a \neq 0, 1$

If $a = 2.71828 \ldots = e$, we write

$f(x) = \log_e x \equiv \ln x$

usually called logarithm to the natural base.

3. **Trigonometric Functions**

These are the circular functions

sinx, cos x and tanx.

There are also their reciprocals

$$\sec x = \frac{1}{\cos x}, \quad \operatorname{cosec} x = \frac{1}{\sin x} \text{ and } \cot x = \frac{1}{\tan x} \quad .$$

Furthermore, $\tan x = \frac{\sin x}{\cos x}$ and so $\cot x = \frac{\cos x}{\sin x}$.

The following results hold:

(a) $\sin^2 x + \cos^2 x = 1$ – Pythagoras' theorem

(b) $1 + \tan^2 x = \sec^2 x$

(a) $1 + \cot^2 x = \operatorname{cosec}^2 x$

(b) $\sin (x + y) = \sin x \cos y + \cos x \sin y$

(c) $\cos (x + y) = \cos x \cos y - \sin x \sin y$

(d) $\tan (x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$

4. Inverse Trigonometric Functions

These are the functions

$f(x) = \sin^{-1} x$ (also called arc sine)

$f(x) = \cos^{-1} x$ (arc cosine).

$f(x) = \tan^{-1} x$ (arc tangent)

$f(x) = \operatorname{cosec}^{-1} x$ (arc cosech x)

$f(x) = \sec^{-1} x$ (arc secant x)

$f(x) = \cot^{-1} x$ (arc cotangent x)

5. Hyperbolic Functions

These are functions defined in terms of the exponential functions as follows:

(a) $\sinh (x)$ or $\sinh x = \frac{1}{2} (e^x - e^{-x})$.

(b) $\cosh x = \frac{1}{2} (e^x + e^{-x})$.

(c) $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

(d) $\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$

(e) $\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$

(f) $\coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

6. **Inverse Hyperbolic Functions**

These include the following:

(a) $\sinh^{-1} x$ (b) $\cosh^{-1} x$ (c) $\tanh^{-1} x$
(d) $\text{cosech}^{-1} x$ (e) $\text{sech}^{-1} x$ (f) $\coth^{-1} x$.

(See notes on hyperbolic functions).

GRAPHS OF FUNCTIONS

Let us now consider the graphs of certain special functions.

(1) **Exponential Functions**

$f(x) = a^x, \quad a \neq 0, 1 \ , \ x \in R$.

For $a > 1$,

When $x = 0, \ a^0 = 1$

As $x \to -\infty$, $a^x \to 0^+$

As $x \to +\infty$, $a^x \to \infty$

$f(x) = a^{-x}$, $a > 1$, $x \in R$.

When $x = 0, \ a^0 = 1$

As $x \to -\infty$, $a^x \to +\infty$

As $x \to +\infty$, $a^x \to 0^+$

The two graphs are as shown below on the same axes.

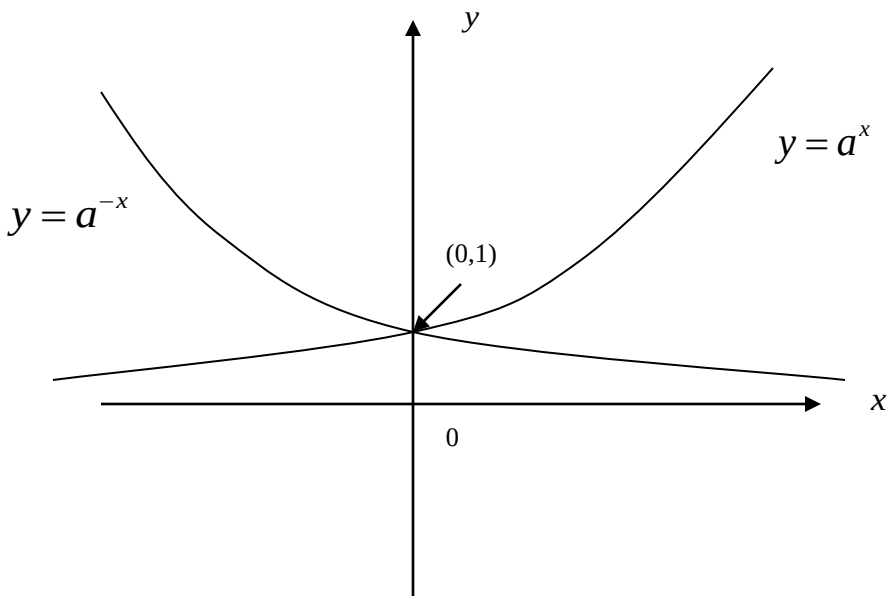


Figure 4.7.1

The exponential function defined by $f(x) = e^x$, where $e \cong 2.72$ is a special exponential function. The number e is called the natural base. How the number e was arrived at will be discussed later.

(2) **Logarithmic Functions**

$f(x) = \log_a x$, $a > 1$, $x \in R$,

<u>When x = *a*, f(*a*) = 1</u>

When $x = 1$, $f(1) = 0$.

As $x \to 0^+$, $f(x) \to -\infty$.

x cannot assume – ve real numbers.

$\text{Log}_a x$ is also the inverse function of the exponential function $y = ax$.

The graph is shown below

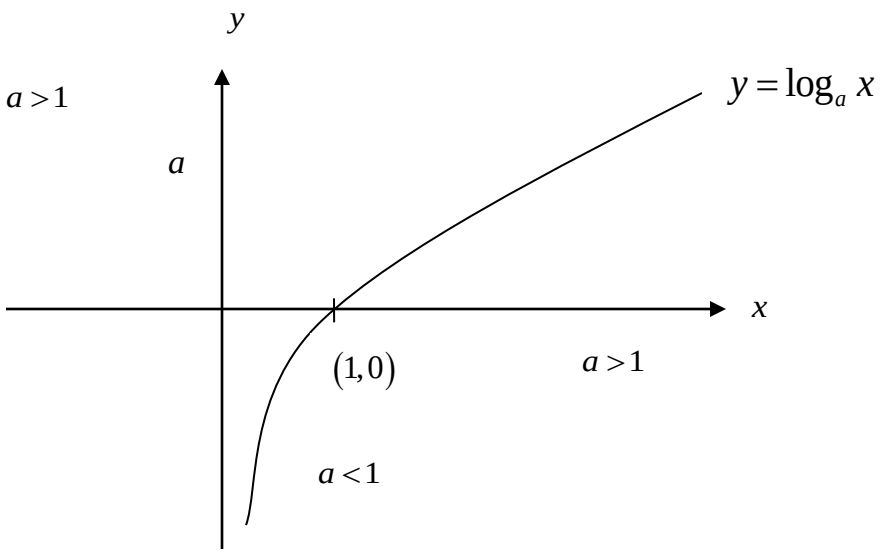


Figure 4.7.2

When a = 10, then $\log_{10} x$ is called the **common logarithm**.

Relationship with the exponential function $y = a^x$, $a > 1$.

The functions $y = ax$ and $y = \log_a x$ are mutually inverse functions. The two graphs are represented on the same graph.

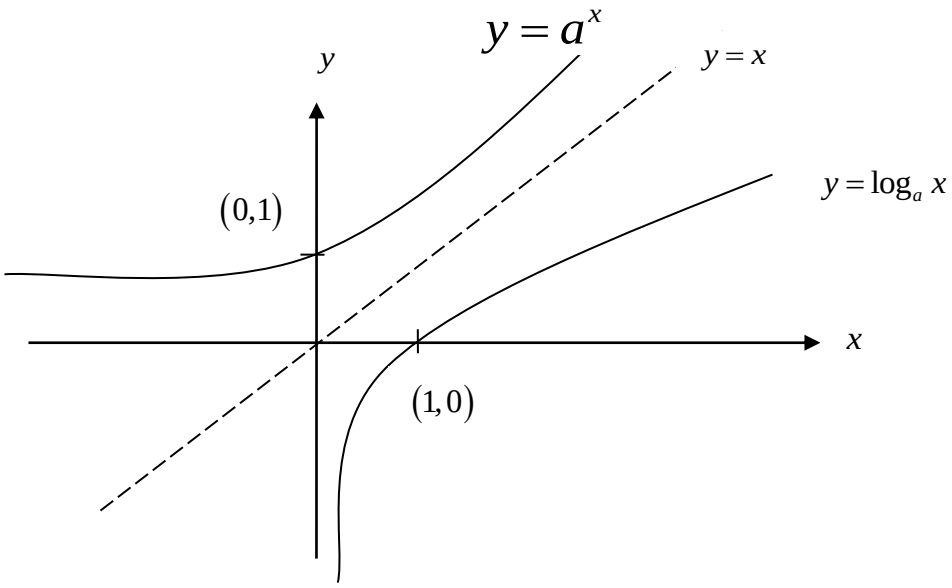


Figure 4.7.3

4.8 EVEN, ODD AND PERIODIC FUNCTIONS

A function f(x) is said to be an **even** function if

$$f(-x) = f(x),\ x \in \Re \qquad \text{----------} \qquad (1)$$

A function is called **odd** if

$$f(-x) = -f(x),\ x \in \Re \qquad \text{----------} \qquad (2)$$

You may be asked to "determine algebraically" whether a function is even or odd. To do this, you take the function and substitute $-x$ in for x, and then simplify. If you end up with the exact same function that you started with (that is, if $f(-x) = f(x)$, so all of the signs are the same), then the function is even. If you end up with the exact opposite of what you started with (that is, if $f(-x) = -f(x)$, so all of the "plus" signs become "minus" signs, and vice versa), then the function is odd. In all other cases, the function is "neither even nor odd".

Example 1

Determine algebraically whether $f(x) = -3x^2 + 4$ is even, odd, or neither.

$$\begin{aligned} f(-x) &= -3(-x)^2 + 4 \\ &= -3(x^2) + 4 \\ &= -3x^2 + 4 \\ \\ &= f(x) \end{aligned}$$

Which means that $f(x)$ is even.

Example 2

Determine algebraically whether $f(x) = 2x^3 - 4x$ is even, odd, or neither.

Solution

$$\begin{aligned} f(-x) &= 2(-x)^3 - 4(-x) \\ &= 2(-x^3) + 4x \\ &= -2x^3 + 4x \end{aligned}$$

The final expression is the exact opposite of the function I started with, by which I mean that the sign on each term has been changed to its opposite, just as if I'd multiplied through by –1:

$$\begin{aligned} -f(x) &= -1[f(x)] \\ &= -[2x^3 - 4x] \\ &= -2x^3 + 4x \end{aligned}$$

This means that $f(x)$ is odd.

Example 3

Determine algebraically whether $f(x) = 2x^3 - 3x^2 - 4x+4$ is even, odd, or neither.

Solution

$$\begin{aligned} f(-x) &= 2(-x)^3 - 3(-x)^2 - 4(-x) + 4 \\ &= 2(-x^3) - 3(x^2) + 4x + 4 \\ &= -2x^3 - 3x^2 + 4x + 4 \end{aligned}$$

This is neither the same function I started with (namely, $2x^3 - 3x^2 - 4x + 4$) nor the exact opposite of what I started with (namely, $-2x^3 + 3x^2 + 4x - 4$). This means that $f(x)$ is neither even nor odd.

Remarks

1) There are functions which are neither even nor odd, and many not periodic.
E.g. $f(x) = x^2 + x^3$

2) Every function f(x) may be expressed as the sum of an even and an odd function, since

$$f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\}.$$

Even odd.

E.g. For $f(x) = (x + 1)^2$.
$f(-x) = (-x + 1)^2$
$\therefore f(x) + f(-x) = (x + 1)^2 + (1 - x)^2$ is an even function; while
$f(x) - f(-x) = (x + 1)^2 - (1 - x)^2$ is an odd function.

Special Even and odd functions

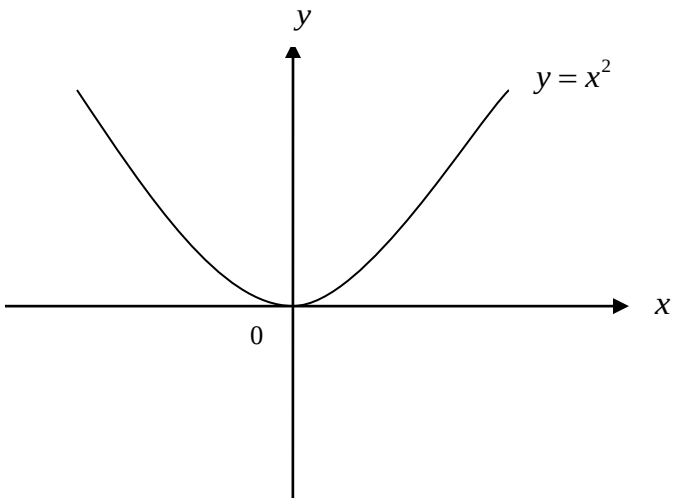
(a) $f(x) = \sin x$ is an odd function since $\sin(-x) = -\sin x$.
(b) $f(x) = \cos x$ is an even function since, $\cos(-x) = \cos x$.

Symmetry

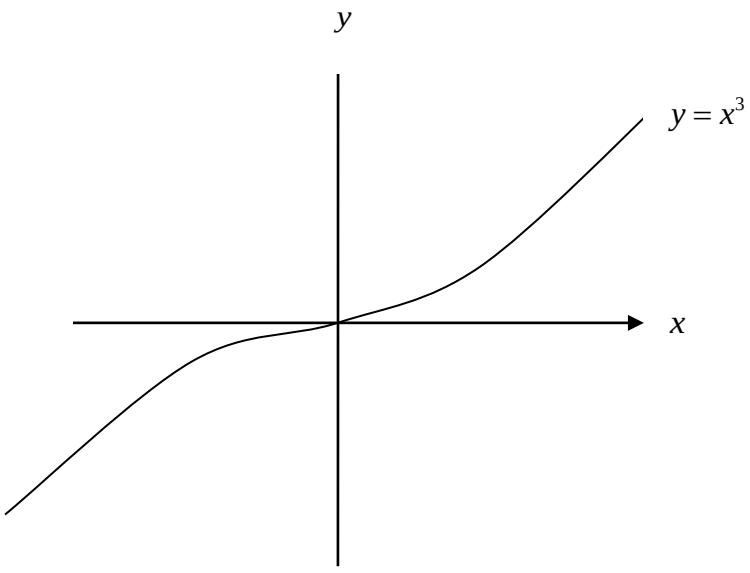
An even function is symmetrical about the y-axis.
An odd function has a rotational symmetry of order 2 about the origin.

Examples

(1) $f(x) = x^2$ is an even function
$f(-x) = (-x)^2 = x^2 = f(x)$



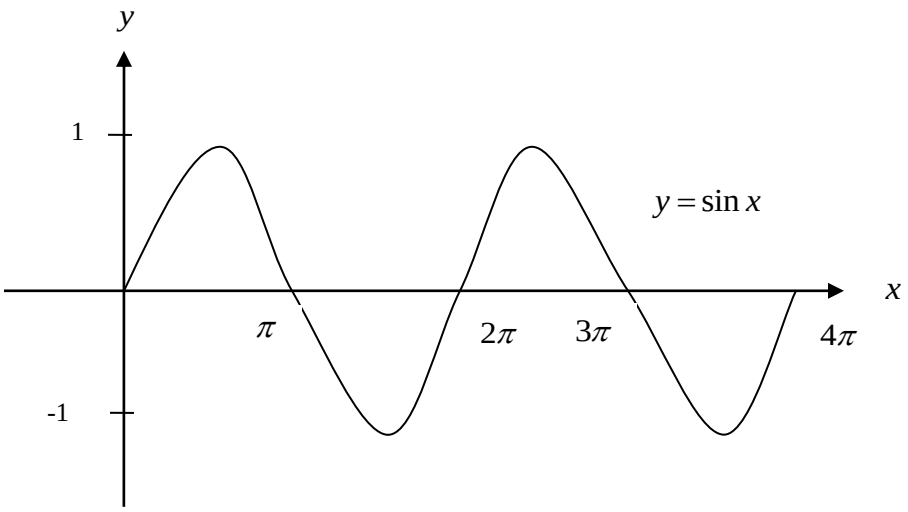
(2) $f(x) = x^3$ is an odd function:
$f(-x) = (-x)^3 = (-1)^3 x^3 = -x^3 = -f(x)$



<u>A function f(x) is said to be a periodic function if ∃ a real number T such that</u>

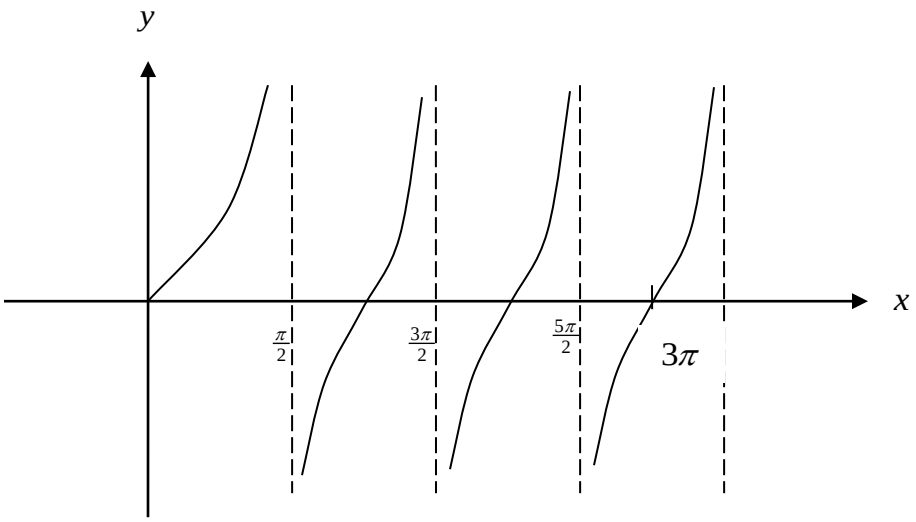
$f(x + T) = f(x)$, $x \in \Re$ ---------- (3)

(3) (a) $f(x) = \sin x$ is periodic with period 2π
$f(x + 2\pi) = \sin(x + 2\pi) = \sin x = f(x)$



(b) f(x) = tan x is periodic with period π

f(x + π) = tan (x + π) = tan x = f(x) .



Note: A periodic function repeats its value after every value x = T.

EXERCISE 4

(1) Consider the function

$$f(x) = \frac{x}{x+1}$$

in the interval [0, 1]. Investigate the following properties of the function f.

(a) Boundedness.
(b) Monotonicity.

(2) Consider the function

$$f(x) = x + \frac{1}{x}$$

in the interval [1, 3]. Describe the following properties of the function:

(e) Boundedness
(f) Monotonicity

(3) Show that

(a) $f(x) = x\sin x$ is an even function
(b) $f(x) = x\cos x$ is an odd function.

(4) From the function

$$f(x) = (x-1)^2,$$

construct

(a) an even function
(b) an odd function

CHAPTER FIVE
LIMITS

5.1 LIMITS OF FUNCTIONS

5.1.1 Definition

Let f(x) be a function defined in a deleted δ - neighbourhood of the point $x = x_0$. The function f(x) is said to have the limit l as x approaches x_0 if f(x) can be made as close to l as we please by choosing x sufficiently close to x_0.

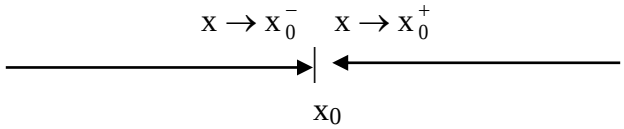
We write

$$\lim_{x \to x_0} f(x) = l . \qquad \text{----------} \qquad (1)$$

In a rigorous language we say that $\lim_{x \to x_0} f(x) = l$ if for every $\in > 0$ (however small, $\exists\ \delta > 0$ ($\delta = \delta(\in)$) such that $|f(x) - l| < \in$, whenever $0 < |x - x_0| < \delta$.

5.2 Right Hand /Left Hand Limits

The lim f(x) as $x \to x_0$ usually depends on how x approaches x_0. Since x_0 is a real number, x can approach x_0 in only two possible directions. (See the figure).



Different results could be obtained for each of the cases. Let us consider the following examples:

1) $\lim_{x \to 0^-} (2x + 1) = 1$

$\lim_{x \to 0^+} (2x + 1) = 1$

2) $\lim_{x \to 0^-} \left(\frac{1}{x}\right) = -\infty$

$\lim_{x \to 0^+} \left(\frac{1}{x}\right) = +\infty$

3) $\lim_{x \to 0^-} \tan^{-1}\left(\frac{1}{x}\right) = -\frac{\pi}{2}$

$\lim_{x \to 0^+} \tan^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2}$

4) Let f(x) be defined by

$$f(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

(See Fig. 1)

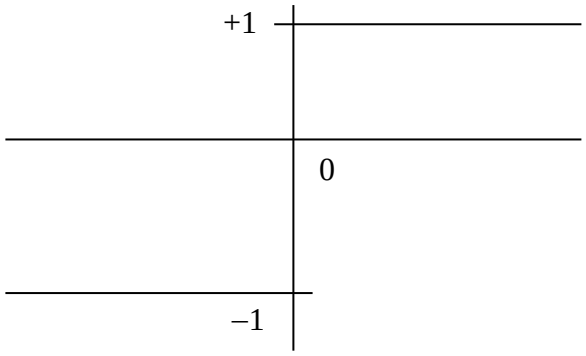


Fig. 1

$$\lim_{x \to 0^-} f(x) = -1$$

$$\lim_{x \to 0^+} f(x) = +1$$

The above examples illustrate the fact that $\lim_{x \to x_0^-} f(x) \neq \lim_{x \to x_0^+} f(x)$ (in general).

Let $\lim_{x \to x_0^-} f(x) = l_1$, and

$$\lim_{x \to x_0^+} f(x) = l_2 .$$

Then l_1 is called the left hand limit, while l_2 is called the right hand limit of f(x) as x approaches x_0.

In general $l_1 \neq l_2$. However, if $l_1 = l_2 = l$ then

$$\lim_{x \to x_0} f(x) = l . \qquad \text{----------} \qquad (2)$$

(3) Define

$$f(x) = \begin{cases} \dfrac{|x-1|}{x-1}, & x \neq 1 \\ 0, & x = 1 \end{cases}$$

(i) Draw the graph

(ii) (ii) show that $\lim_{x \to 1} f(x)$ does not exist.

Solution

Graph : for $x < 1$, $|x-1| = -(x-1)$

$\Rightarrow f(x) = -\dfrac{(x-1)}{x-1} = -1$

for $x > 1$, $|x-1| = x-1$

$\Rightarrow f(x) = \dfrac{x-1}{x-1} = 1$

$f(1) = 0$.

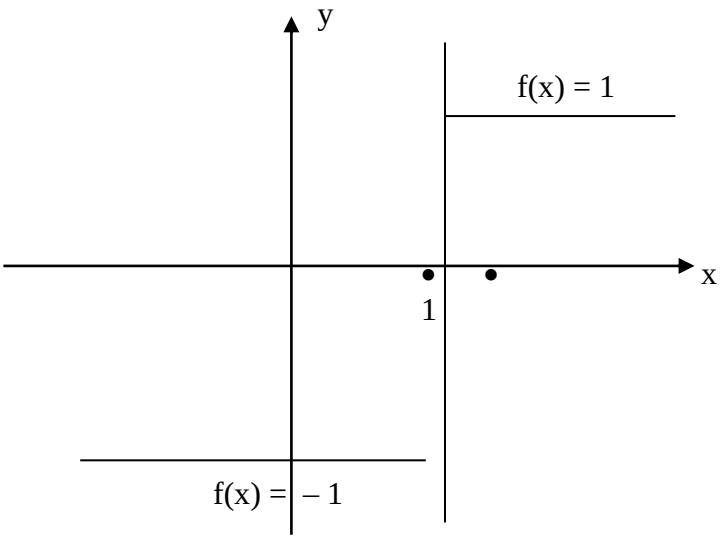


Fig. 2

(ii) $\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} -\dfrac{(x-1)}{x-1} = \lim_{x \to 1^-} (-1) = -1$

$\lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} \dfrac{x-1}{x-1} = \lim_{x \to 1^+} (+1) = 1$

$\therefore \lim_{x \to 1^-} f(x) \neq \lim_{x \to 1^+} f(x)$

$\therefore \lim_{x \to 1} f(x)$ does not exist.

5.3 THEOREMS ON LIMITS

Let

$\lim_{x \to x_0} f(x) = A, \qquad \lim_{x \to x_0} g(x) = B$, where A and B are finite constants.

Then

(a) $\lim_{x \to x_0} (f(x) + g(x)) = \lim_{x \to x_0} f(x) + \lim_{x \to x_0} g(x)$

$= A + B$

(b) $\lim_{x \to x_0} (f(x) - g(x)) = \lim_{x \to x_0} f(x) - \lim_{x \to x_0} g(x)$

$= A - B$

(c) $\lim_{x \to x_0} (f(x) \cdot g(x)) = \lim_{x \to x_0} f(x) \cdot \lim_{x \to x_0} g(x)$

$= A \cdot B$

(d) $\lim_{x \to x_0} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \to x_0} f(x)}{\lim_{x \to x_0} g(x)} = \frac{A}{B}$.

provided that $B \neq o$.

(i) If $A \neq o$ and $B = o$, $\lim_{x \to x_0} \left(\frac{f(x)}{g(x)} \right)$ does not exist.

(ii) If $A = 0$ and $B = 0$, the form $\frac{0}{0}$ is indeterminate and the limit may or may not exist.

5.4 Evaluation of Limits

The following rules will assist us when evaluating limits of functions

(1) If f(x) is a polynomial in x, then

$\lim_{x \to x_0} f(x) = f(x_0)$.

E.g. $\lim_{x \to 1} (x^2 + 2x - 5) = 1^2 + 2(1) - 5 = -2$.

(2) If f(x) is an elementary function, then

$\lim_{x \to x_0} f(x) = f(x_0)$.

E.g. $\lim_{x \to 0} \sin x = \sin 0 = 0.$

(3) If $f(x) = \dfrac{(x - x_0)^{\alpha}\ g(x)}{(x - x_0)^{\alpha}\ h(x)}$, $x \neq x_0$ and α is real, then

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} \frac{g(x)}{h(x)} .$$

E.g. For $f(x) = \dfrac{x^3 - 8}{x - 2}$, $x \neq 2$

$$\lim_{x \to 2} f(x) = \lim_{x \to 2} \frac{x^3 - 8}{x - 2} = \lim_{x \to 2} \frac{(x - 2)(x^2 + 2x + 4)}{x - 2}$$

$$= \lim_{x \to 2} (x^2 + 2x + 4)$$

$$= 2^2 + 2(2) + 4$$
$$= \underline{\underline{12}}$$

(2) $\lim_{x \to \infty} \left(1 + \dfrac{1}{x}\right)^{x} = e$

(3) $\lim_{x \to 0^+} (1 + x)^{\frac{1}{x}} = e$

5.5 Limit as $x \to \infty$ and as $x \to -\infty$

If $f(x)$ approaches a number L as $x \to \infty$, then we write $\lim_{x \to \infty} = L$.

In similar way, we write $\lim_{x \to -\infty} = L$ if $f(x)$ approaches a number L as $x \to$-∞.

Examples

1. Let $f(x) = \dfrac{1}{x}, x \neq 0$.

   As $x \to \infty$, or as $x \to -\infty$, $f(x)$ takes on values closer and closer to zero.

   Thus $\lim_{x \to \infty} \dfrac{1}{x} = 0 = \lim_{x \to -\infty} \dfrac{1}{x}$.

2. Evaluate $\lim_{x \to \infty} \left\{ \dfrac{3x^2 + 7x - 6}{4x^2 + 3x + 6} \right\}$.

Here, we divide each term by the height degree of the numerator and the denominator, i.e., 2

$$\lim_{x\to\infty}\left\{\frac{3x^2+7x-6}{4x^2+3x+6}\right\}=\lim_{x\to\infty}\left\{\frac{\frac{3x^2}{x^2}+\frac{7x}{x^2}-\frac{6}{x^2}}{\frac{4x^2}{x^2}+\frac{3x}{x^2}+\frac{6}{x^2}}\right\}=\lim_{x\to\infty}\left\{\frac{3+\frac{7}{x}-\frac{6}{x^2}}{4+\frac{3}{x}+\frac{6}{x^2}}\right\}=\frac{3}{4}$$

The greatest integer function truncation or "birthday'' function, indicated by [x], is the function that drops any fractional quantity over an integer. For example, [2.99999] = 2 and [π] = 3.

As an example

$\lim_{x\to 2^+}[x]=2$, but $\lim_{x\to 2^+}[x]=2$.

5.6 L' HOSPITAL'S RULE

Suppose we have to find the limiting value of a function $F(x)=\frac{f(x)}{g(x)}$ at $x=c$, when direct substituting of *g(x,)*

$x=c$ gives the indeterminate form $\frac{0}{0}$ i.e., at $x=c$, $f(x)=0$ and $g(x)=0$.

In this case, the ratio of the differential coefficients of the numerator and denominator at *x* = *c* provided, of course, that both $f'(c)$ and $g'(c)$ are not zero themselves.

$$\therefore \lim_{x\to c}\left\{\frac{f(x)}{g(x)}\right\}=\lim_{x\to c}\left\{\frac{f'(x)}{g'(x)}\right\}$$

This is known as L' HOSPITAL'S rule and is extremely useful for finding limiting values when the differential coefficients of the numerator and denominator can easily be found.

Example

Find $\lim_{x\to 1}\left\{\frac{x^3+x^2-x-1}{x^2+2x-3}\right\}$

Solution

$$\lim_{x\to 1}\left\{\frac{x^3+x^2-x-1}{x^2+2x-3}\right\}=\frac{0}{0}$$ (Indeterminate)

$$\lim_{x\to 1}\left\{\frac{3x^2+2x-1}{2x+2}\right\}=\frac{4}{4}=1$$

$$\therefore \quad \lim_{x\to 1}\left\{\frac{x^3+x^2-x-1}{x^2+2x-3}\right\}=1$$

Example

Find $\lim_{x\to 0}\left\{\frac{x^2-\sin 3x}{x^2+4x}\right\}$

Solution

$\lim_{x\to 0}\left\{\frac{x^2-\sin 3x}{x^2+4x}\right\}$ Direct substitution gives $\frac{0}{0}$, so we apply L' HOSPITAL'S rule which gives

$$\lim_{x\to 0}\left\{\frac{2x-3\cos 3x}{2x+4}\right\}=\frac{-3}{4}$$

Example

Determine $\lim_{x\to 0}\left\{\frac{x-\sin x}{x^2}\right\}$

Solution

$$\lim_{x\to 0}\left\{\frac{x-\sin x}{x^2}\right\}=\frac{0}{0}$$ (indeterminate)

Applying L'HOSPITAL'S rule, we have

$$\lim_{x\to 0}\left\{\frac{\sin x}{2x}\right\}=\frac{0}{2}=0$$

5.7 Other Indeterminate Forms

Other indeterminate forms include $\frac{\infty}{\infty}$, ∞ - ∞, and 0.

Examples

1. Evaluate $\lim_{x\to 0}\left\{\frac{1}{x}-\frac{1}{\sin x}\right\}$

This expression is an example of an indeterminate form of the type ∞ - ∞ as x → 0. In order to use L' HOSPITAL'S rule, we first rewrite the expression as $\frac{1}{x}-\frac{1}{\sin x}=\frac{\sin x-x}{x\sin x}$, and this latter expression is of the form $\frac{0}{0}$ as $x\to 0$.

Using L' HOSPITAL'S rule, we obtain $\lim_{x\to 0}\left\{\frac{1}{x}-\frac{1}{\sin x}\right\}=\lim_{x\to 0}\left\{\frac{\sin x-x}{x\sin x}\right\}$

$$=\lim_{x\to 0}\left\{\frac{\cos x-1}{\sin x+x\cos x}\right\}=\frac{0}{0}\quad\text{(Indeterminate)}$$

Applying L' Hospital's rule again, we have $\lim_{x\to 0}\left\{\frac{-\sin x}{2\cos x-x\sin x}\right\}=0$

$$\therefore\quad \lim_{x\to 0}\left\{\frac{1}{x}-\frac{1}{\sin x}\right\}=0$$

2. Evaluate $\lim_{x\to 0^+} x\ln x$. This is of form 0. ∞.

To use L'Hospital's rule we must first write *xlnx* so that it has the form $\frac{0}{0} or \frac{\infty}{\infty}$ as $x\to 0^+$.

$$x\ln x=\frac{\ln x}{\frac{1}{x}}$$

Where both numerator and denominator of the right side become infinite as $x\to 0^+$.
Using L'Hospital's rule ,we obtain

$$\lim_{x\to 0^+} x\ln x=\lim_{x\to 0^+}\left\{\frac{\ln x}{\frac{1}{x}}\right\}=\lim_{x\to 0^+}\left\{\frac{\frac{1}{x}}{\frac{-1}{x^2}}\right\}=\lim_{x\to 0^+}\{-x\}=0.$$

EXERCISE 5

(1) A function f(x) is defined as

$$f(x) = \frac{x^3 + 1}{x + 1}, \quad x \neq -1.$$

Evaluate $\lim_{x \to -1} f(x)$.

(2) Draw the graph of the function

$$f(x) = |x - 2|$$

By determining $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$ show that $\lim_{x \to 2} f(x)$ exists.

(3) Evaluate the following:

(a) $\lim_{x \to -2} \frac{(x + 2)(x^2 + 3)}{x^2 - 4}$

(b) $\lim_{x \to 0} e^x \sin 2x$

(c) $\lim_{x \to 1} \left\{ \frac{1}{x - 1} \left(\frac{1}{x + 3} - \frac{2}{3x + 5} \right) \right\}$

(4) A function f(x) is defined by

$$f(x) = \begin{cases} -1, & x < -1 \\ x & -1 < x < 1 \\ 1, & x > 1 \end{cases}$$

(i) Draw the graph of f(x).

(ii) Show that $\lim_{x \to -1} f(x)$ and $\lim_{x \to 1} f(x)$ both exist.

1. Determine:

a) $\lim_{x\to 0^+}\left\{\frac{1}{x}-\frac{2}{x}\right\}$ b) $\lim_{x\to 0^+}\left\{\frac{1}{x^2}-\frac{1}{x}\right\}$ c) $\lim_{x\to 0^+}\left\{\frac{2}{x}-\left(\frac{2}{x}-5\right)\right\}$

2. a) $\lim_{x\to\infty}\left\{\frac{1+x}{1-x}\right\}$ b) $\lim_{x\to\infty}\left\{\frac{3x^2+4x-5}{2x^3-6x^2+x+1}\right\}$ c) $\lim_{x\to-\infty}\left\{\frac{x^3}{x^3+3}\right\}$

In each of problems 3 through 7, determine whether $\lim_{x\to c^+} f(x)$, $\lim_{x\to c^-} f(x)$, and $\lim_{x\to c} f(x)$ exist. Evaluate each limit that does exist.

3. $f(x)=\begin{cases}\sqrt{1-x^2} & -1<x<1\\ & C=1\\ x, & x>1\end{cases}$

4. $f(x)=\begin{cases}\frac{1}{x^2}, & x<0\\ & C=0\\ x^2, & x>0\end{cases}$

5. $f(x)=\begin{cases}\sec x, & \frac{-\pi}{2}<x<0\\ & C=\frac{\pi}{2}\\ \operatorname{cosec} x, & 0<x<\frac{\pi}{2}\end{cases}$

6. $\begin{cases}\sin x & x<\frac{\pi}{2}\\ 0, & x=\frac{\pi}{2} \quad C=\frac{\pi}{2}\\ \sin 3x, & x>\frac{\pi}{2}\end{cases}$

7. $f(x)=\begin{cases}\sin x, & x<0\\ 1, & x=0 \quad C=0\\ \sin 3x, & x>0\end{cases}$

L'HOSPITAL'S RULE

Find the following:

i.) $\lim_{x\to 0}\dfrac{\sin x}{x}$

ii) $\lim_{x\to 0}\dfrac{1-\cos x}{x^2}$

iii) $\lim_{x\to 1}\dfrac{\ln x}{x^2-1}$

iv) $\lim_{x\to 0}\left\{\dfrac{x+\sin x}{x^2+x}\right\}$

v) $\lim_{x\to 0^+}\left\{\dfrac{\ln x}{\cot x}\right\}$

vi) $\lim_{x\to 0}\left\{\dfrac{\sin x-x}{\tan x-x}\right\}$

vii) $\lim_{x\to 0}\left\{\dfrac{1-\cos^2 x}{x^2}\right\}$

viii) $\lim_{x\to \frac{\pi}{2}}\{\sec x-\tan x\}$

ix) $\lim_{\to 0^+}\left\{\dfrac{1}{x}+\ln x\right\}$

x) $\lim_{x\to 0}\left\{\dfrac{\sin^2 x}{2x^2}\right\}$

CHAPTER 6
CONTINUITY AND DIFFERENTIATION

6.1 CONTINUITY OF A FUNCTION

Definition:

A function $f(x)$ is said to be continuous at a point $x = x_0$ if the following conditions hold:

(i) $\lim_{x \to x_0} f(x)$ exists,

(ii) $f(x_0)$ is defined, and

(iii) $\lim_{x \to x_0} f(x) = f(x_0)$

Any function which fails any one of the above three conditions at x_0 is said to be discontinuous at x_0.

Examples

(i) Given that $f(x) = \dfrac{x^2 - 8}{x - 3}$. Is f continuous at $x = 3$?

Solution

f is not defined at $x=3$.

Hence f is not continous at $x=3$.

(ii) Given that $g(x) = \begin{cases} \dfrac{1}{x}, & \text{if } x \neq 0 \\ 2, & \text{if } x = 0 \end{cases}$. Is g continious at $x = 0$?

Solution

g fails to be continuous at $x = 0$, since $\lim_{x \to 0} g(x)$ does not exist.

(iii) Given that $h(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & \text{if } x \neq 3 \\ 7, & \text{if } x = 3 \end{cases}$. Is h continious at $x = 3$?

Solution

$\lim_{x\to 3} h(x) = 6$ and $h(3) = 7$, therefore $\lim_{x\to 3} h(x) \neq h(3)$. Hence h is not continious at $x = 3$

(iv) Given that $h(x) = \begin{cases} \dfrac{x^2 - 9}{x - 3}, & \text{if } x \neq 3 \\ 6, & \text{if } x = 3 \end{cases}$. Is h continious at $x = 3$?

Solution

$\lim_{x\to 3} h(x) = 6$ and $h(3) = 6$, therefore $\lim_{x\to 3} h(x) = h(3)$. Hence h is continious at $x = 3$

Equivalently we could say that $f(x)$ is continuous at the point $x = x_0$ if $f(x_0)$ is defined and $\lim_{x\to x_0} f(x) = f(x_0)$

Using the $\in - \delta$ definition we say that f(x) is continuous at $x = x_0$ if given $\in > o$ (however small) $\exists\ \delta > o$ such that $|f(x) - f(x_0)| < \in$, whenever $|x - x_0| < \delta$.

Right Hand/Left Hand Continuity

If only $\lim_{x\to x_0^-} f(x) = f(x_0)$, where $f(x_0)$ is defined, then f(x) is continuous on the left; if only $\lim_{x\to x_0^+} f(x) = f(x_0)$, then f(x) is continuous on the right.

Example

Given that $f(x) = \begin{cases} \sqrt{x^2}, & \text{if } x \geq -3 \\ -3, & \text{if } x < -3 \end{cases}$. Is f continious at $x = -3$?

Solution

$\lim_{x\to -3^+} f(x) = 3$

$\lim_{x\to -3^-} f(x) = -3$

$\lim_{x\to -3^+} f(x) \neq \lim_{x\to -3^-} f(x)$. Therefore $\lim_{x\to -3} f(x)$ doesnot exist.

Hence f is not continious at $x = -3$

Continuity in an Interval

A function f(x) is said to be continuous in an interval (a, b) if f(x) is defined for every $x_0 \in (a, b)$ and $\lim_{x\to x_0} f(x) = f(x_0)$.

For the closed interval [a, b], f(x) is continuous in [a, b] if f(x) is continuous in (a, b) and in addition

$\lim_{x \to a^+} f(x) = f(a)$ and $\lim_{x \to b^-} f(x) = f(b)$, where $f(a)$ and $f(b)$ are defined.

THEOREMS ON CONTINUITY

1. If f(x) and g(x) are both continuous at a point $x = x_0$, then $f(x) \pm g(x)$, $f(x) \cdot g(x)$ and $\frac{f(x)}{g(x)}$ are all continuous at $x = x_0$ (provided that $g(x_0) \neq 0$ in the last case).

2. Every polynomial in x, $P_n(x)$ is continuous for all real x.

3. If $y = f(x)$ is continuous at $x = x_0$ and $z = g(y)$ is continuous at $y = y_0$ and if $y_0 = f(x_0)$, then the function $z = g\{f(x)\}$, called the composite or function of a function, is continuous at $x = x_0$. That is, a continuous function of a continuous function is continuous.

4. If f(x) is continuous in a closed interval, then it is bounded in the interval.

6.7 DIFFERENTIABILITY

Let f(x) be defined at some point x_0 in the interval (a, b). Then, the derivative of f(x) at x_0 denoted by $f'(x_0)$ is defined as

$$f'(x_0) = \lim_{x \to x_0} \left\{ \frac{f(x) - f(x_0)}{x - x_0} \right\}, \text{ if the rhs limit exists.}$$

Setting $h = x - x_0$, the above definition becomes

$$f'(x_0) = \lim_{h \to o} \left\{ \frac{f(x_0 + h) - f(x_0)}{h} \right\} \quad \text{----------} \quad (4)$$

A function f(x) is said to be **differentiable** at a point x_0 if $f'(x_0)$ exists.

Right Hand/Left Hand Differentiability

A function f(x) is said to be **left hand differentiable** at $x = x_0$ if

$$f'_-(x_0) = \lim_{h \to o^-} \left\{ \frac{f(x_0 + h) - f(x_0)}{h} \right\} \text{exists. ----------} \quad (5)$$

It is **right hand differentiable** at $x = x_0$ if

$$f'_+(x_0) = \lim_{h \to o^+} \left\{ \frac{f(x_0 + h) - f(x_0)}{h} \right\} \text{exists.} \quad \text{----------(6)}$$

The function f(x) is said to be differentiable at $x = x_0$ if and only if $f'_-(x_0) = f'_+(x_0)$.

Differentiability in an Interval

A function f(x) is differentiable in an interval (a, b) if $f'(x_0)$ exists $\forall\ x_0$ in (a, b). In particular, for the closed interval $[a, b]$ $f'(x_0)$ must exist $\forall\ x_0 \in (a, b)$ and $f'_+(a)$ and $f'_-(b)$ must exist.

Examples

(1) Show that the function $f(x) = x^2$ is differentiable at $x = 2$.

Solution

$$f'(2) = \lim_{h \to 0} \left\{ \frac{f(2 + h) - f(2)}{h} \right\}$$

i.e.

$$f'(2) = \lim_{h \to 0} \left\{ \frac{(2 + h)^2 - 2^2}{h} \right\}$$

$$= \lim_{h \to 0} \left\{ \frac{4 + 4h + h^2 - 4}{h} \right\} = \lim_{h \to 0} \left\{ \frac{4h + h^2}{h} \right\}$$

$$= \lim_{h \to 0} \{4 + h\}$$

$$= 4.$$

$\therefore$ $f'(2)$ exists and so f(x) is differentiable at $x = 2$.

(2) Let

$$f(x) = \begin{cases} x^2, & x > 1 \\ 1, & x = 1 \\ x, & x < 1 \end{cases}$$

Find (a) $f'_-(1)$

(b) $f'_+(1)$

Is f(x) differentiable at $x = 1$?

Solution

$$f'_{-}(1) = \lim_{h \to o^{-}} \frac{f(1+h) - f(1)}{h} = \lim_{h \to o^{-}} \frac{(1+h) - 1}{h}$$

$$= \lim_{h \to o^{-}} \frac{h}{h} = \lim_{h \to o^{-}} 1 = 1$$

$$f'_{+}(1) = \lim_{h \to o^{+}} \frac{f(1+h) - f(1)}{h} = \lim_{h \to o^{+}} \frac{(1+h)^2 - 1}{h}$$

$$= \lim_{h \to o^{+}} \frac{2h + h^2}{h}$$

$$= \lim_{h \to o^{+}} (2 + h) = 2$$

Clearly then, $f'_{-}(1) \neq f'_{+}(1)$ $\therefore$ $f'(1)$ does not exist, and so f(x) is not differentiable at $x = 1$.

THEOREM

If a function f(x) is differentiable at a point $x = x_0$, then it is continuous there. The converse, however, is not true.

Proof:

$$f(x_0 + h) - f(x_0) = \left\{ \frac{f(x_0 + h) - f(x_0)}{h} \right\} . \; h$$

$\Rightarrow$

$$\lim_{h \to 0} \left\{ f(x_0 + h) - f(x_0) \right\} = \lim_{h \to 0} \left\{ \frac{f(x_0 + h) - f(x_0)}{h} \right\} . \lim_{h \to 0} h$$

$$= f'(x_0) \; . \lim_{h \to 0} h$$

$$= 0 \text{ , since } \quad f'(x_0) \text{ exists.}$$

$\therefore \lim_{h \to 0} f(x_0 + h) = f(x_0)$

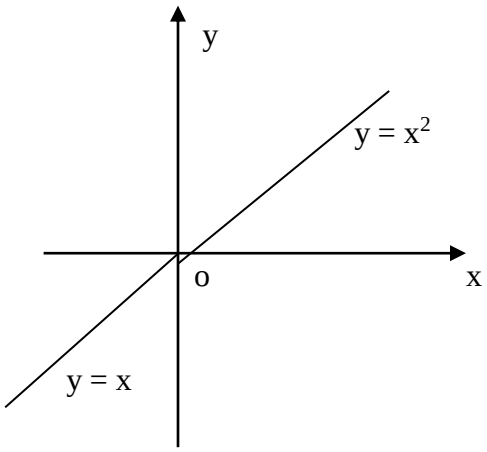
Setting $h = x - x_0$, we have

$$\lim_{x \to x_0} f(x) = f(x_0)$$

which completes the proof.

The converse of this theorem is false. Consider the counter example

$$f(x) = \begin{cases} x^2 & , \quad x > 0 \\ x & , \quad x < 0 \\ 0 & , \quad x = 0 \end{cases}$$



Graph of y = f(x).

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} x = 0$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} x^2 = 0$$

$$f(o) = 0$$

$\therefore$ f(x) is continuous at x = 0.

Now

$$f'_-(o) = \lim_{h \to 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0^-} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0^-} \frac{h - 0}{h}$$

$$= 1$$

$$f'_+(0) = \lim_{h \to 0^+} \frac{f(h) - f(0)}{h}$$

$$= \lim_{h \to 0^+} \frac{h^2 - 0}{h} = \lim_{n \to 0^+} h = 0$$

$\therefore f'_-(0) \neq f'_+(0)$

$\therefore$ f(x) is not differentiable at x = 0.

6.8 DIFFERENTIATION

The derivative of a function f(x) at any point x is defined by

$$f'(x) = \lim_{h \to 0} \left\{ \frac{f(x+h) - f(x)}{h} \right\} \quad .$$

The function $f'(x)$ is also called the Derived Function of f(x).
The process of finding the derivative is called **differentiation**. The derived function $f'(x)$ enables one to obtain the derivative of f(x) at any point $x = x_0$. The following cases will be considered.

1. $f(x) = x^n$, n rational.

(i) for $f(x) = x$

$$f'(x) = \lim_{h \to 0} \left\{ \frac{f(x+h) - f(x)}{h} \right\} \quad .$$

$$= \lim_{h \to 0} \left\{ \frac{(x+h) - x}{h} \right\} = \lim_{h \to 0} \left\{ \frac{h}{h} \right\} = 1 \text{ ----------} \qquad (7)$$

(ii) $f(x) = x^2$

$$f'(x) = \lim_{h \to 0} \left\{ \frac{(x+h)^2 - x^2}{h} \right\} = \lim_{h \to 0} \left\{ \frac{2hx + h^2}{h} \right\} \quad .$$

$$= \lim_{h \to 0} \{2x + h\} = 2x \text{ -----} \quad (8)$$

(iii) $f(x) = x^3$

$$f'(x) = \lim_{h \to 0} \left\{ \frac{(x+h)^3 - x^3}{h} \right\}$$

$$= \lim_{h \to 0} \left\{ \frac{3x^2h + 3xh^2 + h^3}{h} \right\}$$

$$= \lim_{h \to 0} \{3x^2 + 3xh + h^2\} = 3x^2 \text{ ----------} \qquad (9)$$

(iv) $f(x) = x^n$, n integer.

$$f'(x) = \lim_{h\to 0}\left\{\frac{(x+h)^n - x^n}{h}\right\}$$

$$= \lim_{h\to 0}\left\{\frac{x^n + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \ldots + \binom{n}{n-1}xh^{n-1} + h^n - x^n}{h}\right\}$$

$$= \lim_{h\to 0}\left\{\left(\binom{n}{1}x^{n-1} + \binom{n}{2}x^{n-2}h + \ldots + \binom{n}{n-1}xh^{n-2} + h^{n-1}\right)\right\}$$

$$= \binom{n}{1}x^{n-1} = \frac{n!}{(n-1)!\,1!}x^{n-1} = n\,x^{n-1} \qquad \text{----------} \qquad (10)$$

(v) $f(x) = x^{-1} = \frac{1}{x}$

$$f'(x) = \lim_{h\to 0}\left\{\frac{\frac{1}{x+h} - \frac{1}{x}}{h}\right\}$$

$$= \lim_{h\to 0}\left\{\frac{x - (x+h)}{h\,x\,(x+h)}\right\} = \lim_{h\to 0}\left\{\frac{-h}{h(x^2 + xh)}\right\}$$

$$= \lim_{h\to 0}\left\{\frac{-1}{x^2 + xh}\right\} = -\frac{1}{x^2}\ . \text{----- (11)}$$

The same result as for (iv) holds for negative integers.
In fact for all rational numbers n.

$f(x) = x^n$

$\Rightarrow$ $f'(x) = n\,x^{n-1}$

2. **Trigonometric Functions.**

(i) $f(x) = \sin x.$

$$f'(x) = \lim_{h\to 0}\left\{\frac{\sin(x+h) - \sin x}{h}\right\}$$

Using the fact that $\sin A - \sin B = 2\cos\frac{A+B}{2}\cdot\sin\frac{A-B}{2}$ we have

$$f'(x) = \lim_{h\to 0}\left\{\frac{2\cos\frac{(x+h+x)}{2}\ \sin\frac{(x+h-x)}{2}}{h}\right\}.$$

$$= \lim_{h\to 0}\left\{\frac{2\cos\frac{(2x+h)}{2}\ \sin\frac{h}{2}}{h}\right\}$$

$$= \lim_{h\to 0}\cos\left(x+\frac{h}{2}\right)\cdot\lim_{h\to 0}\frac{\sin\frac{h}{2}}{\frac{h}{2}} = \cos x.\ 1$$

$$= \cos x \qquad \text{----------} \qquad (12)$$

(ii) $f(x) = \cos x$

$$f'(x) = \lim_{h\to 0}\left\{\frac{f(x+h) - f(x)}{h}\right\}$$

$$= \lim_{h\to 0}\left\{\frac{\cos(x+h) - \cos x}{h}\right\}$$

$$= \lim_{h\to 0}\left\{\frac{-2\sin\frac{(x+h+x)}{2}\ \sin\frac{(x+h-x)}{2}}{h}\right\}$$

$$= \lim_{h\to 0}\left\{\frac{-2\sin\left(x+\frac{h}{2}\right)\ \sin\frac{h}{2}}{h}\right\}$$

$$= \lim_{h\to 0}\left\{-\sin\left(x+\frac{h}{2}\right)\right\}\cdot\lim_{h\to 0}\left\{\frac{\sin\frac{h}{2}}{\frac{h}{2}}\right\}$$

$$= -\sin x\ .\ 1 = \underline{\underline{-\sin x}}\ . \qquad \text{----------} \qquad (13)$$

3. **Exponential Function**

Differentiation of $y = a^x$, $a > 1$

$$f'(x) = \frac{dy}{dx} = \lim_{h\to 0}\frac{a^{x+h} - a^x}{h}$$

$$= \left(\lim_{h\to 0}\frac{a^h - 1}{h}\right)a^x = k.a^x, \qquad \text{----------} \qquad (14)$$

where

$$k = \lim_{h\to 0}\left(\frac{a^h - 1}{h}\right) \qquad \text{----------} \qquad (15)$$

Note that when $x = 0$, for $a \neq 0$, $\frac{dy}{dx} = k$. This is the gradient of the tangent to the curve $y = a^x$ at $x = 0$. In order to determine the value of k, let us attempt to find $\frac{dy}{dx}$ by an alternative approach:

If $y = a^x$

Taking log on both sides, we have

$$\log_e y = x \log_e a$$

$$\Rightarrow \quad \frac{1}{y}\frac{dy}{dx} = \log_e a$$

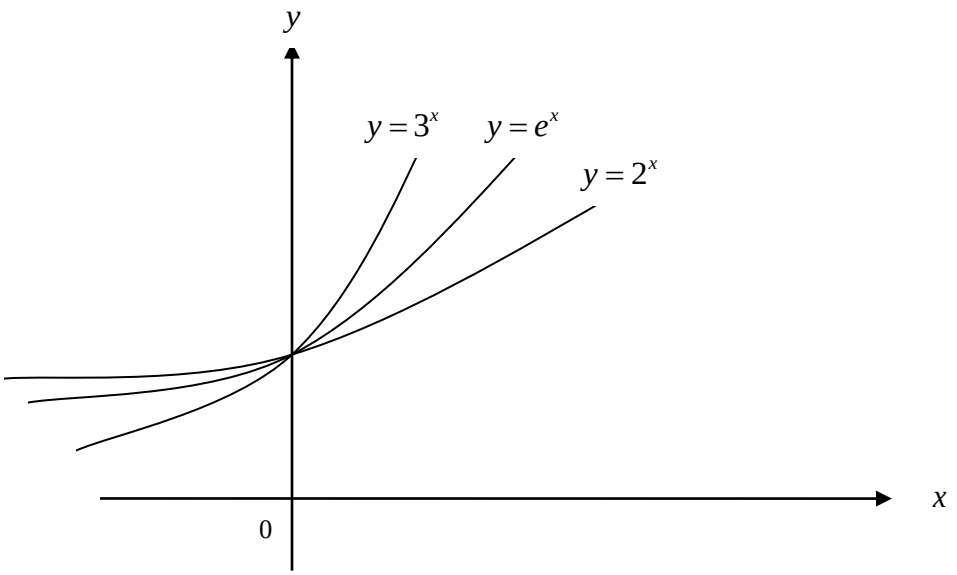
$$\Rightarrow \quad \frac{dy}{dx} = y \log_e a = a^x . \log_e a \qquad \text{----------} \qquad (16)$$

Comparing this result with 14 above, we have $\lim_{h\to 0}\left\{\frac{a^h - 1}{h}\right\} = \log_e a$.

Thus $y = a^x$

$\Rightarrow$ $\boxed{\frac{dy}{dx} = a^x \log_e a}$ ---------- (17)

Graphs of 2^x, e^x, 3^x



For $a > 1$, $\frac{dy}{dx}$ at $x = o$ increases. The evidence from the graph shows that $T_2 < T_3$, where T_2, T_3 are the tangents of the curves for 2^x and 3^x respectively. Gradient of $T_2 < 1$ while grad of $T_3 > 1$.

Let $y = e^x$, $2 < e < 3$ s.t. gradient of $T_e = 1$.

Hence for $a = e$, $k = 1$,

$y = e^x$,

and $\frac{dy}{dx} = e^x$.

We have obtained an important result that

$\boxed{y = e^x \Rightarrow \frac{dy}{dx} = e^x}$ ---------- (18)

4. **Logarithmic Function**

Let $y = \log_e x$. Then

$$x = e^y$$

$$\therefore \ 1 = e^y \frac{dy}{dx}$$

$$\therefore \ \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$

i.e. $\boxed{y = \log_e x \Rightarrow \frac{dy}{dx} = \frac{1}{x}}$ ---------- (19)

6.9 RULES FOR DIFFERENTIATION

The following rules are valid and can be proved from the definition of the derivative.

Let f(x) and g(x) be functions. Then the following rules hold:

(i) $h(x) = f(x) \pm g(x)$

$\Rightarrow h'(x) = f'(x) \pm g'(x)$ – **Sum and Difference**

(ii) $h(x) = f(x) \ . g(x)$

$\Rightarrow h'(x) = f'(x) \ . g(x) + f(x) g'(x)$ – **Product Rule**

(iii) $h(x) = \frac{f(x)}{g(x)}$

$\Rightarrow h'(x) = \frac{g(x) f'(x) - f(x) g'(x)}{[g(x)]^2}$ – **Quotient Rule**

The Chain Rule

Let

$h(x) = f[u(x)]$,

then

$$h'(x) = \frac{df}{du} \times \frac{du}{dx}$$

EXERCISE 6

(1) Define

$$f(x) = \begin{cases} \dfrac{x^3 - 8}{x - 2}, & x \neq 2 \\ 0, & x = 2 \end{cases}$$

Is f(x) continuous at x = 2? If not, how can f(x) be redefined so that it is continuous at x = 2?

(2) Construct a function f(x) which is
- (a) Continuous at x = 0,
- (c) Differentiable at x = 1,
- (d) Continuous at x = 1 but not differentiable there.

(3) Construct a function f(x) which is
- (a) Continuous at x = 0,
- (c) Differentiable at x = 1,
- (d) Continuous at x = 1 but not differentiable there.

(4) Differentiate the following functions w.r.t x:
- (a) $f(x) = (x^2 + 1)^3$
- (b) $f(x) = \tan x$
- (c) $f(x) = x^2 \sin x$
- (d) $f(x) = \cos (2x + 1)^2$
- (e) $f(x) = e^{x^2+x}$.

CHAPTER SEVEN

ROLLE'S THEOREM AND THE MEAN-VALUE THEOREMS

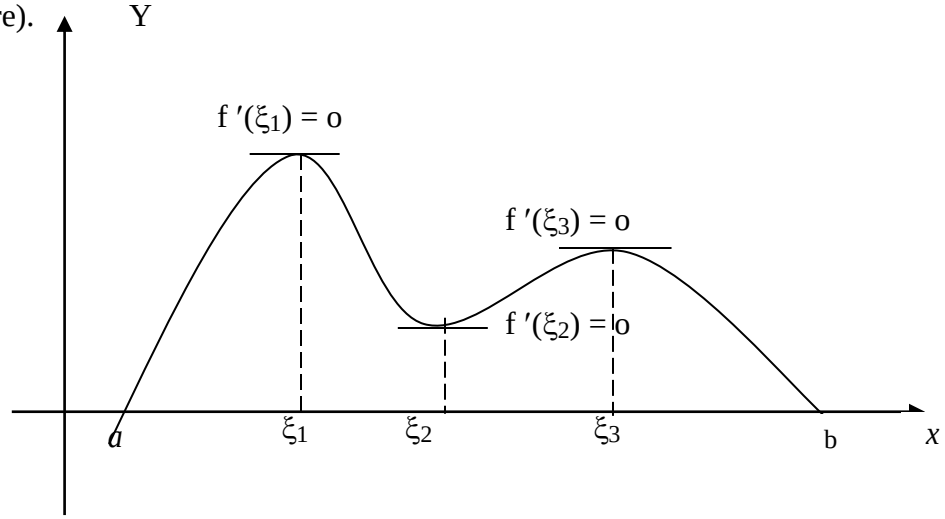
In this chapter, Rolle's Theorem will be studied, proved and used to discuss what is commonly known as the mean-value theorems.

7.1 ROLLE'S THEOREM

If a function f(x) is continuous in a closed interval $[a, b]$ and differentiable in an open interval (a, b), and if $f(a) = f(b) = 0$, then $\exists$ at least one value of $x = \xi \in (a, b)$ such that $f'(\xi) = 0$.

Geometrical Interpretation

If the curve $y = f(x)$ is continuous in $[a, b]$ and has a unique tangent at every point in $[a, b]$ except possibly at $x = a$ and $x = b$, then there will always be at least one tangent to the curve whose gradient is zero (i.e. parallel to the x-axis).
(see the figure).



Proof:

If $f(x) = 0$, $\forall x \in [a, b]$, then $f'(x) = o$ $\forall x$ in (a, b) and the theorem is trivially true.

Suppose now that $f(x) \neq o$ for some x in $[a, b]$.

Since $f(x)$ is continuous in $[a, b]$, $\exists$ m, M s.t.

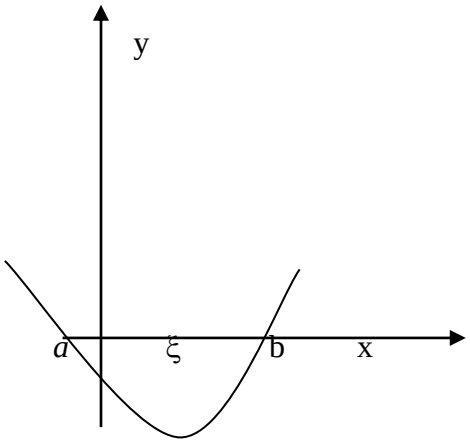
M = minimum f(x), and M = maximum f(x).

Since $f(x) \neq o$, M and m are not simultaneously zero. Suppose $M \neq 0$ and that

$f(\xi) = M$ for some $\xi \in (a, b)$.

Then for any h,

$f(\xi + h) \leq f(\xi)$.



If $h > 0$, then

$$\frac{f(\xi + h) - f(\xi)}{h} \leq 0$$

$$\Rightarrow \quad \lim_{h \to 0^+} \left\{ \frac{f(\xi + h) - f(\xi)}{h} \right\} \leq 0$$

If $h < 0$, then

$$\frac{f(\xi + h) - f(\xi)}{h} \geq 0$$

$$\Rightarrow \quad \lim_{h \to 0^-} \left\{ \frac{f(\xi + h) - f(\xi)}{h} \right\} \geq 0$$

Since $f'(x)$ exists $\forall\, x \in (a, b)$, in particular $f'(\xi)$ exists

$\therefore\ f'_-(\xi) = f'_+(\xi)$

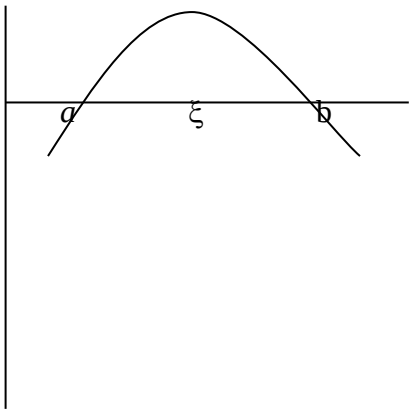
$$\Rightarrow \quad \lim_{h \to 0^-} \left\{ \frac{f(\xi + h) - f(\xi)}{h} \right\} = \lim_{h \to 0^+} \left\{ \frac{f(\xi + h) - f(\xi)}{h} \right\} = 0.$$

$$\Rightarrow f'(\xi) = 0.$$

On the other hand if $m \neq o$, and that $f(\xi) = m$ for some $\xi \in (a, b)$.

Then for any h,

$f(\xi + h) \geq f(\xi)$.



If $h > o$, then

$$\frac{f(\xi + h) - f(\xi)}{h} \geq 0$$

$$\Rightarrow \quad \lim_{h \to 0^+} \left\{ \frac{f(\xi + h) - f(\xi)}{h} \right\} \geq 0$$

If $h < 0$, $\dfrac{f(\xi + h) - f(\xi)}{h} \leq 0$

$$\Rightarrow \quad \lim_{h \to 0^-} \left\{ \frac{f(\xi + h) - f(\xi)}{h} \right\} \leq 0$$

Since $f'(\xi)$ exists, we have $f'_-(\xi) = f'_+(\xi) = 0$

$\therefore f'(\xi) = 0.$

This completes the proof of Rolles Theorem.

Example

Verify Rolle's Theorem for the case where

$$f(x) = x(2x - 1), \qquad [0, \tfrac{1}{2}].$$

Solution

$f(x)$ is continuous $\forall$ x real and so for $[0, \frac{1}{2}]$.

$f'(x) = 4x - 1$; and so $f'(x)$ exists in $(0, \frac{1}{2})$.

$f(0) = f(\frac{1}{2}) = 0.$

$\therefore$ All the conditions of Rolle's Theorem are satisfied

$\therefore \exists\, x = \xi \in (0, \frac{1}{2})$ s.t. $f'(\xi) = 4\xi - 1 = 0$

$\Rightarrow \xi = \frac{1}{4} \in (0, \frac{1}{2}).$

Remarks

The condition $f(a) = f(b) = 0$ in Rolle's Theorem may be replaced with the condition $f(a) = f(b)$.

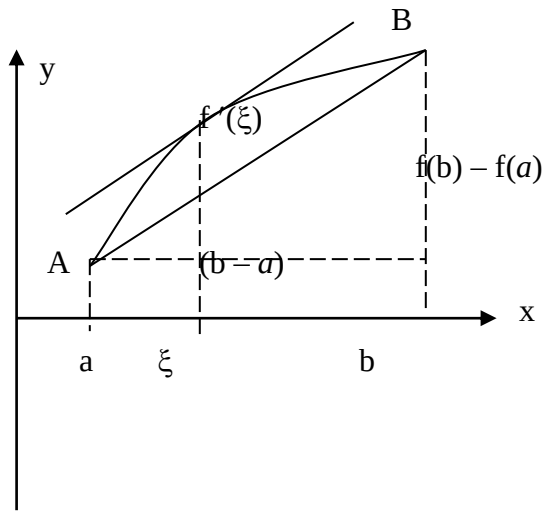
7.2 THE FIRST MEAN-VALUE THEOREM

It states that if f(x) is continuous in the interval $[a, b]$ and differentiable in (a, b) then $\exists$ at least one value of $x = \xi \in (a, b)$ s.t.

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

Geometrical Interpretation

There is a tangent parallel to the chord AB. (See figure)



Proof

Construct the function

$$G(x) = f(x) - y$$

where y is the equation of the chord AB joining two points A and B on the curve $y = f(x)$ from $x = a$ to $x = b$.

Equation of chord AB is

$$y - f(a) = \frac{f(b) - f(a)}{b - a}\ (x - a)$$

$$\Rightarrow \quad y = \frac{f(b) - f(a)}{b - a}\ (x - a) + f(a).$$

$$\therefore \quad G(x) = f(x) - \frac{f(b) - f(a)}{b - a}\ (x - a) - f(a).$$

It is clear that G(x) is continuous and differentiable in $[a, b]$, since f(x) is.

Furthermore,

$$G(a) = f(a) - 0 - f(a) = 0,$$

$$G(b) = f(b) - \frac{f(b) - f(a)}{b - a}\ (b - a) - f(a), \quad b \neq a$$

$$= f(b) - f(b) + f(a) - f(a) = 0.$$

Thus G(x) satisfies all the conditions of Rolle's theorem. Therefore $\exists\ x = \xi \in (a, b)$ s.t. $G'(\xi) = 0$.

Now $G'(x) = f'(x) - \dfrac{f(b) - f(a)}{b - a}$

$\therefore\ G'(\xi) = 0 \Rightarrow f'(\xi) - \dfrac{f(b) - f(a)}{b - a} = 0$

$\Rightarrow f'(\xi) = \dfrac{f(b) - f(a)}{b - a}$.

Example

Verify the first mean-value theorem for the function

$f(x) = (x^2 - 3x + 1)$, $[0, 1]$.

Solution

$f(x)$ is continuous in $[0, 1]$ and differentiable in $(0,1)$, since $f(x)$ is a polynomial.

$f'(\xi) = 2\xi - 3.$

$$\frac{f(1) - f(0)}{1 - 0} = \frac{1^2 - 3(1) + 1 - 1}{1} = -2$$

$\therefore\ f'(\xi) = 2\xi - 3 = -2$

$\Rightarrow 2\xi = 1$

$\Rightarrow \xi = \frac{1}{2} \in (0, 1).$

which verifies the theorem.

Remarks

If $f(b) = f(a)$, the theorem reduces to Rolle's theorem. Thus Rolle's Theorem is a special case of the first mean-value theorem.

7.3 THE SECOND MEAN-VALUE THEOREM

(OR CAUCHY'S GENERALIZED MEAN-VALUE THEOREM)

This theorem states that if $f(x)$ and $g(x)$ are continuous in $[a, b]$ and differentiable in (a, b), then there exists at least one value of $x = \xi \in (a, b)$ such that

$$\frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)},$$ $g(a) \neq g(b)$ and where $f'(x)$ and $g'(x)$ are not simultaneously zero.

Proof:

Construct the function

$H(x) = f(x) - f(a) - \alpha \{g(x) - g(a)\}$, where α is a constant.

$H(x)$ is continuous in $[a, b]$, since $f(x)$ and $g(x)$ are also continuous in $[a, b]$.

$H(x)$ is differentiable in (a, b) since $f(x)$ and $g(x)$ are also differentiable in (a, b).

Now,

$$H(a) = f(a) - f(a) - \alpha \{g(a) - g(a)\} = 0$$

$$H(b) = f(b) - f(a) - \alpha \{g(b) - g(a)\} = 0,$$

provided that $\alpha = \dfrac{f(b) - f(a)}{g(b) - g(a)}$.

If we suppose that α takes on this value, then there exists at least one value of $\xi \in (a, b)$ s.t. $H'(\xi) = 0$.

$$H'(x) = f'(x) - \alpha g'(x)$$

$$H'(\xi) = o \Rightarrow f'(\xi) - \alpha g'(\xi) = 0$$

$$\Rightarrow \alpha = f'(\xi) / g'(\xi)$$

$$\therefore \quad \frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)} .$$

This completes the proof.

Remarks:

Note that the 2nd Mean-Value theorem reduces to the 1st Mean-Value theorem if g(x) =x.

7.4 TAYLOR'S THEOREM OF THE MEAN

If the nth derivative $f^{(n)}(x)$ of $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) then there exists at least one value of $x = \xi \in (a, b)$ such that

$$f(b) = f(a) + \frac{f'(a)}{1!}(b - a) + \frac{f''(a)}{2!}(b - a)^2 + \ldots$$

$$+ \frac{f^{(n)}(a)}{n!}(b - a)^n + Rn ,$$

where Rn, called the remainder, may be written in one of the forms

$$Rn = \frac{f^{(n+1)}(\xi)}{(n+1)!}(b-a)^{n+1} \quad \ldots\ldots\ldots \quad \text{Lagrange's form}$$

$$Rn = \frac{f^{(n+1)}(\xi)(b-\xi)^n(b-a)}{n!} \quad \ldots\ldots\ldots \quad \text{Cauchy's form.}$$

Remarks:

1) Lagrange's form of the remainder will be generally used.

2) Consider the sub-interval $[x_0, x]$ in (a, b), then Taylor's theorem may be written as

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \ldots$$

$$+ \frac{f^n(x_0)}{n!}(x-x_0)^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1} .$$

This last form of Taylor's theorem is also known as Taylor's series for $f(x)$ with a remainder.

3) Taylor's theorem is used to approximate $f(x)$ by a polynomial, and Rn is called the error term.

4) If $\lim_{n\to\infty} Rn = 0$, the infinite series

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \ldots$$

is called a Taylor's series for $f(x)$ about $x = x_0$. In the case $x_0 = 0$, the Taylor series obtained is called a Maclaurin Series.

INTEGRATION

9.1 INDEFINITE INTEGRATION

$$\int x^n dx = \frac{x^{n+1}}{(n+1)} + c \qquad (1).$$

If we look at the right-hand side and differentiate the function $\frac{x^{n+1}}{(n+1)}$, we obtain the function x^n; that is, the function on the left hand side of (1). Therefore, it is clear that integration is the reverse of differentiation.

Consider the differential coefficient of the two functions: (1) $\frac{1}{3}x^3 + 7$ *and* (2) $\frac{1}{3}x^3 + 9$.

The result in each case is x^2. Reversing the process, and integrating x^2 produces the result, $\frac{1}{3}x^3$. However, no information is obtained about the constant in the original functions. Therefore, treating integration as the reverse of differentiation produces an indefinite result as far as the constant is concerned. Consequently, a constant of integration *C* is always added to the result. This process is called **indefinite integration.**

For example,

$$\int x^2 dx = \tfrac{1}{3}x^3 + C$$

$$\int 5x^6 dx = 5\int x^6 dx = \tfrac{5}{7}x^7 + C$$

The constant *C* cannot be determined unless some further information is known about the resulting function.

Example

Evaluate $\int x^4 dx$. Determine the constant of integration if the graph of the resulting function passes through the point (1, 3).

Solution

$$\int x^4 dx = \frac{x^5}{5} + C.$$

Now the curve $y = \frac{x^5}{5} + C$ passes through the point (1, 3).

We therefore substitute $x = 1,\ y = 3$ into this equation and obtain

$$3 = \frac{1^5}{5} + C$$

$$= \frac{1}{5} + C$$

Therefore $C = 2\frac{4}{5} = \frac{14}{5}$

Hence $y = \frac{x^5}{5} + \frac{14}{5}$

9.2 Types

TYPE 1: Negative and fractional indices

The general indefinite integral is given by

$$\textbf{RULE} \quad \int px^n dx = p\int x^n dx = \frac{pn^{n+1}}{(n+1)} + C \qquad (1.1)$$

This can apply to any value of *n*, including fraction and negative indices, as shown in the example below. The exception to this rule is when $n = -1$. When $n = -1$, the above rule gives

$\int \frac{1}{x} dx = \int x^{-1} dx = \frac{x^0}{0} + C$, which gives an infinite result.

Example

Determine the following integrals,

$(i)\int x^{-3} dx,$ $(ii)\int x^{\frac{3}{4}} dx$

$(iii)\int \frac{1}{x^5} dx,$ $(iv)\int \frac{1}{\sqrt{x}} dx$

Solution

$$(i)\int x^{-3} dx = \frac{x^{-3+1}}{(-3+1)} + C$$
$$= \frac{x^{-2}}{-2} + C$$

$$(ii)\int x^{\frac{3}{4}} dx = \frac{x^{\frac{3}{4}+1}}{(\frac{3}{4}+1)} + C$$
$$= \frac{x^{\frac{7}{4}}}{\frac{7}{4}} + C$$
$$= \tfrac{4}{7} x^{\frac{7}{4}} + C$$

$(iii)\int \frac{1}{x^5} dx.$

In order to evaluate the integral, the function inside the integral sign must be put in the form x^n in the numerator. Therefore, the integral becomes

$$\int x^{-5}dx = \frac{x^{-5+1}}{(-5+1)} + C$$
$$= \frac{-x^{-4}}{4} + C$$

$$(iv)\int \frac{1}{\sqrt{x}}dx = \int \frac{dx}{x^{\frac{1}{2}}}$$
$$= \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$$
$$= 2x^{\frac{1}{2}} + C$$

TYPE 2: Integration of polynomials

In this type, each term can be integrated separately as shown in the example below.

Example

Determine the following integral, $\int \left(x^2 + 3x + 4\right)dx$.

Each term is considered separately.

$$\int \left(x^2 + 3x + 4\right)dx = \int x^2dx + \int 3xdx + \int 4dx$$
$$= \tfrac{1}{3}x^3 + \tfrac{3}{2}x^2 + 4x + C$$

TYPE 3: Integration of functions reducible to polynomials

Some functions can be integrated by converting them into the form shown in the example above. Two functions, which can be integrated in this way, are shown in the example below,

Example

Determine

$$(a)\int_1^2 (x+3)^2dx$$
$$(b)\int \frac{x^2 + 3x + 4}{x^4}dx$$

Solution

(a) The first step is to expand the bracket, and then integrate term by term.

$$\int_1^2 (x+1)^2\,dx = \int_1^2 (x^2+6x+9)dx$$
$$= \left[\tfrac{1}{3}x^3 + 3x^2 + 9x\right]_1^2$$
$$= \left[\tfrac{8}{3}+12+18\right] - \left[\tfrac{1}{3}+3+9\right]$$
$$= 20\tfrac{1}{3}$$

(b) The first step is to divide x^4 into each term in the numerator, that is

$$\int (x^{-2}+3x^{-3}+4x^{-4})dx = \frac{x^{-1}}{-1} + \frac{3x^{-2}}{-2} + \frac{4x^{-3}}{-3} + C$$
$$= -x^{-1} - \frac{3}{2}x^{-2} - \frac{4}{3}x^{-3} + C$$

9.3 Functions which Integrate to give the Logarithm Function

$\frac{d}{dx}(\log_e x) = \frac{1}{x}$. Reversing the procedure, $\int \frac{1}{x}dx = \ln|x| + C$.

If we look at the integral, it is obvious that the numerator, 1, is an exact differential of the denominator. Whenever this characteristic is present in an integral, the result is the logarithm of the denominator, even when the denominator is complex.

RULE: In an integral, if the numerator is the differential coefficient of the denominator apart from a multiplying constant, the integral may be carried out by substituting *u* for the denominator.

Example

Determine $\int \frac{(2x+1)dx}{2x^2+2x+7}$.

The differential coefficient of the denominator is $4x+2$, which is the same as the numerator, apart from a multiplying constant.

Let $u = 2x^2+2x+7;$ $\quad \frac{du}{dx} = 4x+2 = 2(2x+1)$

$$du = 2(2x+1)dx$$
$$(2x+1)dx = \tfrac{1}{2}du$$

The integral becomes

$$\int \frac{\frac{1}{2}du}{u} = \tfrac{1}{2}\int \frac{du}{u}$$
$$= \tfrac{1}{2}\ln u + C$$
$$= \tfrac{1}{2}\ln\left|2x^2+2x+7\right| + C$$

It is possible to write down the integral without having to go through the process of substitution each time. In the above example, the numerator is half the differential coefficient of the denominator, (which is 4*x* + 2). If we multiply the numerator by two, then it becomes the exact differential of the denominator. However, in order to keep the complete integral unchanged, if we multiply the inside by 2, we must multiply the outside by $\frac{1}{2}$. Thus, the integral becomes

$\frac{1}{2}\int \frac{2(2x+1)dx}{2x^2+2x+1}$.

The actual integral is the logarithm of the denominator, with the multiplying factor $\frac{1}{2}$; that is,

$\frac{1}{2}\ln\left|2x^2+2x+7\right|+C$.

Example

$$\int \frac{1}{3x+2}dx$$

Solution

$$\int \frac{1}{3x+2}dx = \frac{1}{3}\int \frac{3}{3x+2}dx = \frac{1}{3}\ln\left|3x+2\right| + C$$

9.4 Integration of Trigonometric Functions

$$(a)\int \sin x dx = -\cos x + C$$
$$(b)\int \cos x dx = \sin x + C$$
$$(c)\int \tan x dx$$

$\tan x$ is written as $\frac{\sin x}{\cos x}$ so that the integral is $\int \frac{\sin x dx}{\cos x}$.

The differential coefficient of cos$x$ is –sin$x$, which is equal to the numerator, apart from a multiplying constant of –1.

Let $u = \cos x$,

Then $\frac{du}{dx} = -\sin x$

$du = -\sin x dx$

that is $\sin x dx = -du$

The integral is $\int \frac{-du}{u} = -\ln|u| + C$

$= -\ln|\cos x| + C$

$$\int \tan x dx = -\ln|\cos x| + C$$

(c) $\cot x dx$ is obtained in a similar manner to give

$$\int \cot x dx = \ln|\sin x| + C$$

9.5 Integration of the Squared Trigonometric Functions

As the reverse of differentiation, we have

(a) $\boxed{\int \sec^2 x dx = \tan x + C}$

(b) $\boxed{\int \cos ec^2 x dx = -\cot x + C}$

Using these results, it is possible to integrate

(c) $\int \tan^2 x dx = \int (\sec^2 x - 1) dx$,

Using result (a) above, we have

$$\boxed{\int \tan^2 x dx = \tan x - x + C}$$

(d) $\int \cot^2 x dx = \int (\cos ec^2 x - 1) dx$.

Using result (b) above, we have

$$\boxed{\int \cot^2 x dx = -\cot x - x + C}$$

(e) $\int \sin^2 x dx$.

In order to carry out the integration, $\sin^2 x$ is expressed in terms of $\cos 2x$, i.e.

$$\cos 2x = 2\cos^2 x - 1$$

$$\cos^2 x = \tfrac{1}{2} + \tfrac{1}{2}\cos 2x$$

$$\int \cos^2 x dx = \int (\tfrac{1}{2} + \tfrac{1}{2}\cos 2x) dx$$

which gives

$$\boxed{\int \cos^2 x dx = \tfrac{1}{2}x + \tfrac{1}{4}\sin 2x + C}$$

Example

Determine $\int \sin 4x . \sin 2x dx$.

Expressing the product as sums or differences,

$$\frac{C+D}{2} = 4x, \qquad \frac{C-D}{2} = 2x$$

That is $\quad C + D = 8x, \qquad C - D = 4x$

Solving the two equations in terms of x,

$C = 6x$

$D = 2x$

From standard equation, $\text{-2}\sin\frac{\text{C}+\text{D}}{2}\sin\frac{C-D}{2} = \cos C - \sin D$

So that, $\sin 4x.\sin 2x = \frac{1}{2}\left[\cos 2x - \cos 6x\right]$

Therefore, $\int \sin 4x.\sin 2x dx = \int \frac{1}{2}\left[\cos 2x - \cos 6x\right] dx$

$$= \frac{\sin 2x}{4} - \frac{\sin 6x}{12} + C$$

9.6 Integration of the Exponential Function

It was shown that $\frac{d}{dx}(e^x) = e^x$.

Reversing the process, we have $\int e^x dx = e^x + C$.

When the exponential contains a compound exponent, such as $\int e^{-3x+1} dx$, the integral may be a substitution method.

Let $u = -3x + 1$, so that $\frac{du}{dx} = -3$

Then $du = \frac{du}{dx} dx$

So that in this case $du = -3dx$

Therefore $dx = -\frac{1}{3} du$

Substituting, in terms of *u*, the integral becomes

$$\int e^{-3x+1} dx = \int -\frac{1}{3} e^u du$$

$$= -\frac{1}{3} e^u + C$$

$$= -\frac{1}{3} e^{-3x+1} + C$$

Example

Evaluate $\int_1^2 e^{2x+1} dx$.

Solution

Using the above rule in the integral becomes

$$\left[\frac{e^{2x+1}}{2}\right]_1^2 = \frac{1}{2}e^5 - \frac{1}{2}e^3$$
$$= \tfrac{1}{2}(148.4) - \tfrac{1}{2}(20.1)$$
$$= 64.15$$

EXERCISE 9

Obtain the following integrals

1. $\int x\left\{\frac{1}{\sqrt{x}}-1\right\}^2 dx$.

2. $\int_0^1 \sqrt{x\left(1-x^3\right)^2}\,dx$.

3. $\int_1^2 \left(x^3-3x^2+5\right)dx$.

4. $\int\left(x+\frac{1}{x}\right)^2 dx$

5. $\int_4^9\left(\sqrt{x}+\frac{1}{\sqrt{x}}+2\right)dx$

6. $\int\left(\sqrt{x^5}-\sqrt[5]{x}\right)dx$

7. $\int_1^4 \sqrt{x}\left(x^2-1\right)dx$

8. Simplify (i) $\int\left(r^7-r^6\right)dr$

(ii) $\int\frac{y^5-y^3}{y^2}dy$

EXERCISE 13

1. $\int e^{3x}dx$

2. $\int e^{-2x-1}dx$

3. $\int e^{-\frac{1}{2}x}dx$

4. $\int e^{-5\theta+\frac{1}{3}}d\theta$

5. $\int_1^2 e^{-t+1}dt$

6. $\int_{-1}^1 e^{\frac{1}{2}x-2}dx$

CHAPTER TEN

CO-ORDINATE GEOMETRY

5.0 **Introduction:** Students are already familiar with topics in co-ordinate geometry such as Equations of straight lines and circles.

In this chapter we shall concentrate on loci, which define plane shapes we generally call CONIC SECTIONS. Treatment will be carried out in the Cartesian rectangular co-ordinates. The last section of this chapter will deal with polar curves – curves whose equations are presented in the polar co-ordinate system.

5.1 CONIC SECTIONS

There are three basic conic sections – the Parabola, the Ellipse and the Hyperbola.

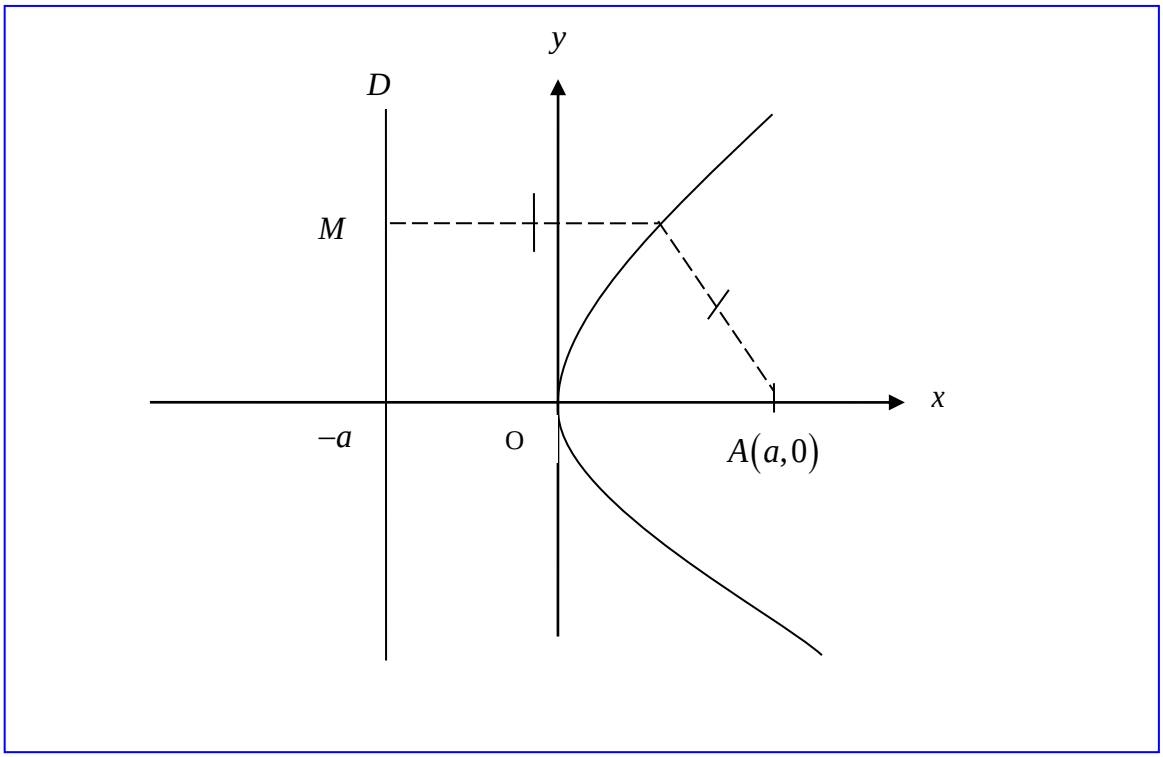
A. THE PARABOLA

Let $P(x, y)$ be a variable point in the $x - y$ plane. If the locus of the point $P(x, y)$ is such that

it is the same distance from a fixed point (called the Focus) and from a fixed line (called the

Directrix), then the locus of the point P defines a curve called a PARABOLA.

Consider a point A (a, o) on the x-axis and a line D with equation $x = -a$ (parallel to the y-axis). A (a, 0) is the focus while D($x = -a$) is the directrix.



Let M be a point on D such that PM is parallel to the x-axis. Then by the definition of a parabola

$$|PM| = |PA| .$$

That is,

$$(x + a) = \sqrt{(x-a)^2 + (y-o)^2} ,$$

$\Rightarrow$

$$y^2 + (x-a)^2 = (x+a)^2$$

$\Rightarrow$ $$y^2 + x^2 - 2ax + a^2 = x^2 + 2ax + a^2$$

$\Rightarrow$ $$y^2 = 4ax.$$

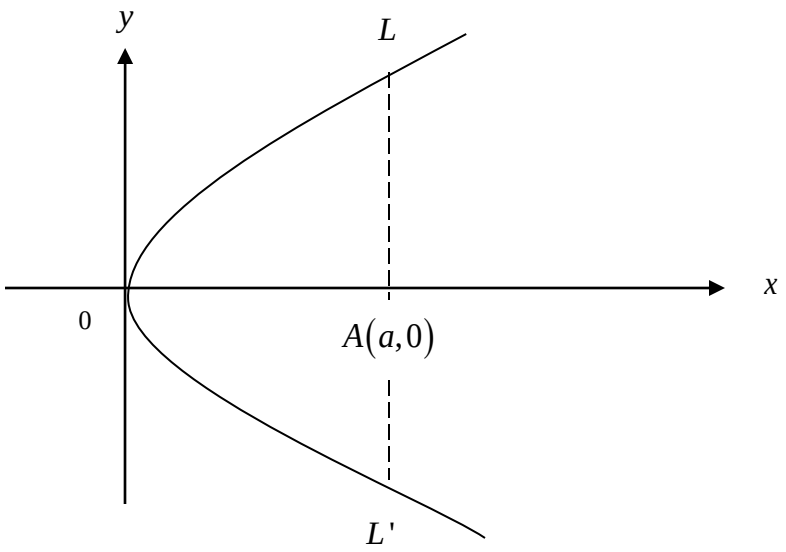
Thus the general equation of a parabola is

$$\boxed{y^2 = 4ax} \qquad \text{----------(1)}$$

This general equation is in its simplest form.

The Graph:

When $x = 0$, $y^2 = 0 \Rightarrow y = 0$ (repeated) thus the curve touches the y-axis at 0; when $y = 0$, $x = 0$. Curve meets the axes only at the origin, 0.



The curve is symmetrical about the x-axis. The line LL′ is called the Latus Rectum of the parabola.

Parametric Equations
The equations

$x = at^2, \quad y = 2at,$ ---------- (2)

where t is a parameter, are called the parametric equations of the parabola. They satisfy the equation of the curve since

$y^2 = (2at)^2 = 4a^2t^2 = 4a(at^2) = 4ax$. ---------- (3)

Any point $(at^2, 2at)$ lies on the curve.

Equations of Tangents and Normals

$y^2 = 4ax$

$\Rightarrow$

$$2y\frac{dy}{dx} = 4a \quad \Rightarrow \quad \frac{dy}{dx} = \frac{2a}{y} \text{ ---------} \quad (4)$$

Thus at any point $(at^2, 2at)$, on the curve the tangent has gradient

$$\frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}. \quad \text{--------} \quad (5)$$

The equation of the tangent at any point 't' is then

$$y - 2at = \frac{1}{t}(x - at^2)$$

$\Rightarrow$

$$y - 2at = \frac{x}{t} - at$$

$$\boxed{ty - x - at^2 = 0}$$

$\Rightarrow$ --------- (6)

The gradient of the normal at $(at^2, 2at)$ is $-t$; the equation is then

$y - 2at = -t(x - at^2)$.

$\Rightarrow$ $\boxed{y + tx - 2at - at^3 = 0 \ .}$ ---------- (7)

Length of the Latus Rectum

When $x = a$

$y^2 = 4a(a) = 4a^2$

$\therefore y = \pm 2a$.

Hence the Latus Rectum has length

$|L\,L'| = 4a$. --------- (8)

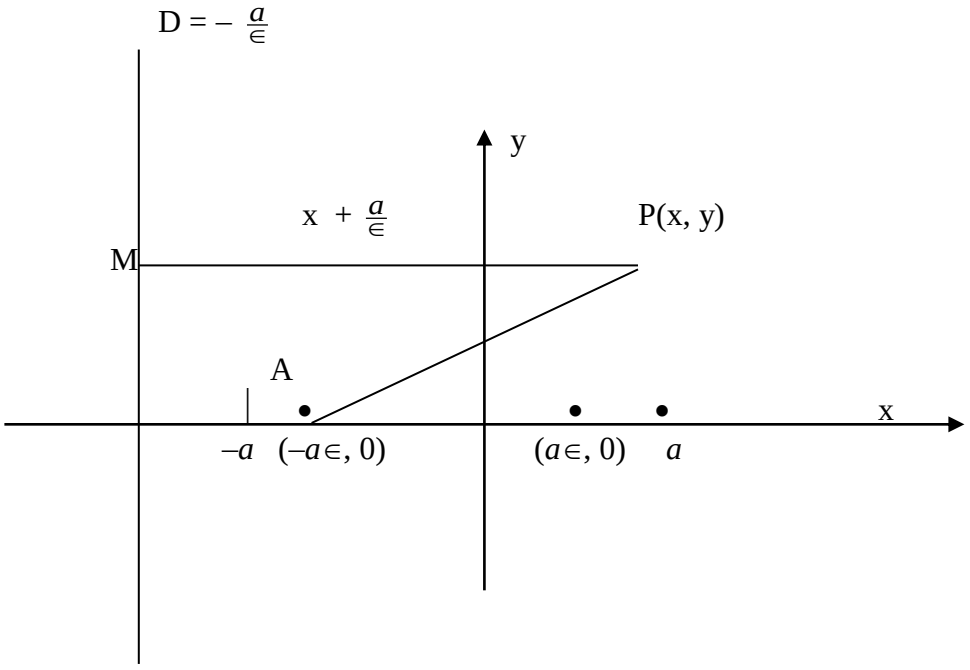
Exercise:

(1) A straight line passing through the focus of the parabola $y^2 = 4ax$ has gradient 1. Determine the points of intersection of the line with the parabola.

(2) The normal to the parabola $y^2 = 4ax$ at the point t $(t \neq 0)$ meets the curve again at the point t'. Find t' in terms of t. Determine a point on the x-axis where the tangent at t' meets the x-axis.

B. **THE ELLIPSE**

The locus of points P(x, y) in a plane such that its distance from a fixed point (the focus) is always less than its distance from a fixed line (the directrix).



From the figure above, with $0 < \in < 1$, and a constant a, the directrix has equation

$$x = -\frac{a}{\in} \quad . \qquad \text{----------} \qquad (9)$$

Then

$$\frac{|PA|}{|PM|} = \in \qquad \text{----------} \qquad (10)$$

$$\Rightarrow \sqrt{(x + a\in)^2 + y^2} = \in \sqrt{\left(x + \frac{a}{\in}\right)^2} .$$

$$\Rightarrow (x + a\in)^2 + y^2 = \in^2 \left(x + \frac{a}{\in}\right)^2 .$$

$$\Rightarrow x^2 + 2a\in x + a^2\in^2 + y^2 = \in^2 \left(x^2 + 2\frac{a}{\in}x + \frac{a^2}{\in^2}\right)$$

$$\Rightarrow x^2(1 - \in^2) + y^2 = a^2(1 - \in^2).$$

Dividing through by $a^2(1 - \in^2)$, we have

$$\frac{x^2}{a^2} + \frac{y^2}{a^2(1 - \in^2)} = 1. \qquad \text{----------} \quad (11)$$

Setting $b^2 = a^2(1 - \in^2)$, we have the equation

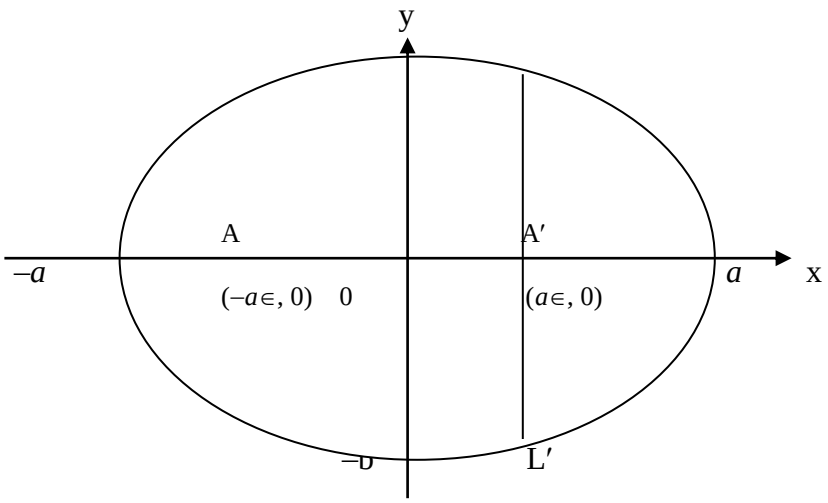
$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1} \qquad \text{----------} \quad (12)$$

This is the equation of the ellipse.

Notes:

1) Since $0 < \in < 1$, $a > b$.

The constant a is called the **semi-major axis** of the ellipse, while b is called its **semi-minor axis**. The ellipse has the shape



2) The number $\in$ is called the **Eccentricity** of the ellipse.
If $\in = 0$, the ellipse reduces to a circle of radius a.

If $\in = 1$, the locus is a parabola.

3) The ellipse is symmetrical in both the x and the y-axes; these are called the **axes** of the ellipse.

4). When $x = a \in$, we have

$$\frac{a^2 \in^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \quad \frac{y^2}{b^2} = 1 - \in^2$$

$$\Rightarrow \quad y^2 = b^2 (1 - \in^2).$$

$$\Rightarrow \quad y = \pm b \sqrt{1 - \in^2} \quad . \qquad \text{----------} \qquad (13)$$

Thus the Latus Rectum of the ellipse has length

$LL' = 2b \ (1 - \in^2)^{½}$.

Equations of Tangents and Normals

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \quad \frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$$

$$\Rightarrow \quad \frac{dy}{dx} = -\frac{b^2 x}{a^2 y} \qquad \text{----------} \qquad (14)$$

At any point (x_1, y_1) on the ellipse, the tangent there has gradient

$$\frac{dy}{dx} = -\frac{b^2}{a^2}\left(\frac{x_1}{y_1}\right) .$$

$\therefore$ The equation of the tangent is

$$y - y_1 = -\frac{b^2}{a^2}\left(\frac{x_1}{y_1}\right)(x - x_1) .$$

$$\Rightarrow \quad a^2 y_1 (y - y_1) = -b^2 x_1 (x - x_1) .$$

$$\Rightarrow \quad a^2 y_1 y - a^2 y_1^2 = -b^2 x_1 x + b^2 x_1^2 .$$

Dividing through by $a^2 b^2$ and re-arranging, we have

$$\frac{y_1 y}{b^2} + \frac{x_1 x}{a^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} \qquad \text{----------} \qquad (15)$$

The rhs equals 1, since (x_1, y_1) lies on the ellipse.
Thus the tangent has equation

$$\boxed{\frac{x_1 x}{a^2} + \frac{y_1 y}{b^2} = 1.} \qquad \text{----------} \quad (16)$$

The gradient of the normal at (x_1, y_1) is $\frac{a^2 y_1}{b^2 x_1}$.

$\therefore$ The normal at (x_1, y_1) has equation

$$y - y_1 = \frac{a^2}{b^2}\frac{y_1}{x_1}(x - x_1).$$

$$\Rightarrow \quad b^2 x_1 (y - y_1) = a^2 y_1 (x - x_1).$$

$$\Rightarrow \quad b^2 x_1 y - b^2 x_1 y_1 = a^2 y_1 x - a^2 y_1 x_1$$

$$\frac{x_1 y}{a^2} - \frac{y_1 x}{b^2} = \frac{x_1 y_1}{a^2} - \frac{x_1 y_1}{b^2}$$

$$\Rightarrow \quad \boxed{\frac{x_1 y}{a^2} - \frac{y_1 x}{b^2} = \left(\frac{1}{a^2} - \frac{1}{b^2}\right) x_1 y_1} \qquad \text{----------} \quad (17)$$

Parametric Equations of the Ellipse.
The equations are

$$x = a\cos t, \quad y = b\sin t, \qquad \text{----------} \quad (18)$$

where t is a parameter. The point (a cost, b sint) lies on the ellipse, since

$$\frac{(a\cos t)^2}{a^2} + \frac{(b\sin t)^2}{b^2} = \cos^2 t + \sin^2 t = 1.$$

The equation of the tangent to the ellipse at any point 't' (ie. (a cost, b sint)) will be

$$\frac{a\cos t}{a^2}x + \frac{b\sin t}{b^2}y = 1,$$

i.e. using equation (5.1.16)

$$\boxed{\left(\frac{\cos t}{a}\right)x + \left(\frac{\sin t}{b}\right)y = 1.} \qquad \text{----------} \quad (19)$$

By a similar procedure, the equation of the normal at any point 't' on the curve can be obtained.

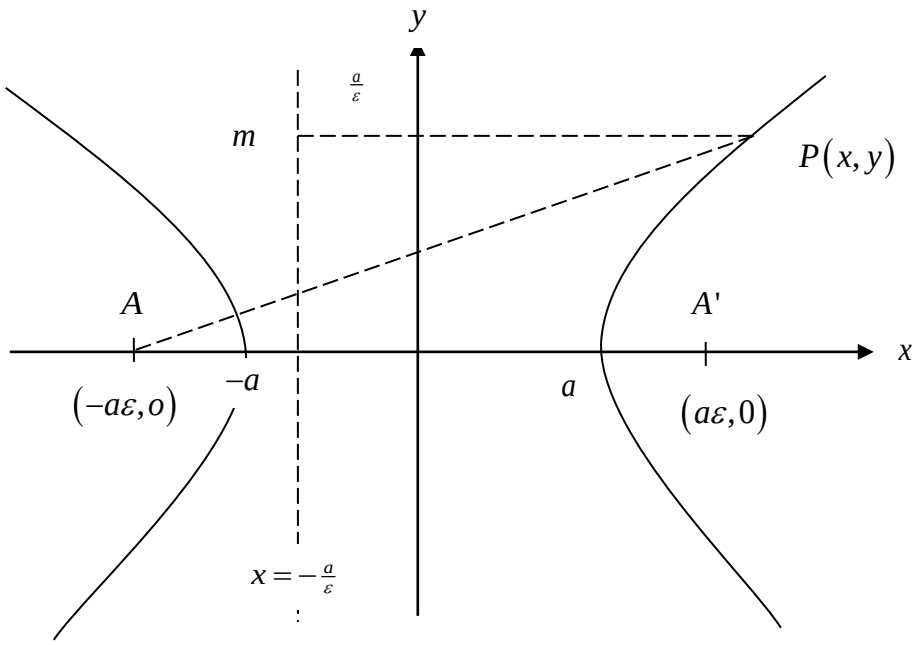
Exercises:

(1) The normal at the point 't' on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the ellipse again at t_1.

Find t_1 in terms of t .

(2) Find the equations of the tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where the Latus rectum meets the curve. Show that the two tangents meet on the x-axis.

C **THE HYPERBOLA**



In the case of the hyperbola, the eccentricity $\in > 1$. Thus by definition

$$\frac{|PA|}{|PM|} > 1 \qquad \text{----------} \qquad (20)$$

i.e.

$$\frac{|PA|}{|PM|} = \in \text{, where } \in > 1.$$

$$\Rightarrow \quad (x + a\in)^2 + y^2 = \in^2 \left(x + \frac{a}{\in} \right)^2 .$$

$$\Rightarrow \quad x^2 + 2a\in x + a^2 \in^2 + y^2 = \in^2 \left(x^2 + 2\frac{a}{\in} x + \frac{a^2}{\in^2} \right) .$$

$$= \in^2 x^2 + 2a\in x + a^2 .$$

Hence

$$x^2 (1 - \in^2) + y^2 = a^2 (1 - \in^2).$$

$\Rightarrow$

$$x^2 (\in^2 - 1) - y^2 = a^2 (\in^2 - 1) \qquad \text{----------} \qquad (21)$$

which leads to the equation

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1} \qquad \text{----------} \qquad (22)$$

where $b^2 = a^2 (\in^2 - 1)$.

Parametric Equations

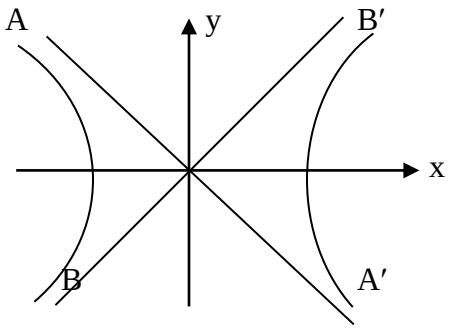
These are

$$x = a \sec t \ , \ y = b \tan t \qquad \text{----------} \qquad (23)$$

or

$$x = a \cosh t \ , \ y = b \sin h\, t. \qquad \text{----------} \qquad (24)$$

Asymptotes



The tangents to the hyperbola at $x = \pm \infty$ are called the asymptotes of the hyperbola.
They have equations

$$y = \pm \frac{b}{a} x \qquad \text{----------} \qquad (25)$$

In the figure, they are shown by the lines AA' and BB'.

When AA' and BB' are perpendicular, then the hyperbola is called a **rectangular hyperbola**. In this case it may be shown that the equation of the hyuperbola reduces to

$$\boxed{xy = c^2,} \qquad \text{----------} \qquad (26)$$

where c is a real constant.

Parametric Equations of the Rectangular Hyperbola.

These are

$$x = ct, \; y = \frac{c}{t} \qquad \text{----------} \qquad (27)$$

Thus, the point 't' has co-ordinates $(ct, \frac{c}{t})$.

Exercise 10

(1) Show that the equation of the tangent to the hyperbola at any point (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.

(2) Find the equations of the tangents and normals to the rectangular hyperbola at any point 't' on it.

(3) The normal to the rectangular hyperbola at the point $(ct, \frac{c}{t})$ meets the hyperbola again at the point 't_1'. Find t_1 in terms of t.